

Pesticide-induced resurgence in brown planthoppers is mediated by action on a suite of genes that promote juvenile hormone biosynthesis and female fecundity

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eLife Assessment

This **useful** manuscript reports mechanisms behind the increase in fecundity in response to sub-lethal doses of pesticides in the crop pest, the brown plant hopper. The authors hypothesize that the pesticide works by inducing the JH titer, which through the JH signaling pathway induces egg development, for which the evidence was judged to be **solid**.

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Abstract

Pesticide-induced resurgence, increases in pest insect populations following pesticide application, is a serious threat to the sustainable control of many highly damaging crop pests. Resurgence can result from pesticide-enhanced pest reproduction, however, the molecular mechanisms mediating this process remain unresolved. Here we show that brown planthopper (BPH) resurgence in rice crops following exposure to sublethal doses of the pesticide emamectin benzoate (EB) results from the coordinated action of a suite of genes that regulate juvenile hormone (JH) levels, resulting in increased JH titer in adult females and enhanced fecundity.

We demonstrate that EB treatment at sublethal levels results in profound changes in female BPH fitness including increased egg maturation and oviposition. This enhanced reproductive fitness results from the EB-mediated upregulation of key genes involved in the regulation of JH, including *JHAMT* and *Kr-h1* and the downregulation of allatostatin (*AstA*) and allatostatin receptor (*AstAR*) expression.

AstA signaling is known to inhibit the production of JH in the corpora allata and hence EB exposure diminishes this inhibitory action. We find that the changes in gene expression

following EB exposure are caused by the allosteric action of this insecticide on its molecular target, the glutamate-gated chloride channel (GluCl α).

Collectively, these results provide mechanistic insights into the regulation of negative pesticide-induced responses in insects and reveal some key actors involved in the JH-signaling pathway that underpin pesticide resurgence.

Significance

Pesticides remain a key means of controlling many of the world's insect pests. However, in some cases, pesticide applications can result in resurgence of pest populations due to pesticide-induced increases in fecundity. In the current study, we show that pesticide resurgence in the brown planthopper (BPH) following exposure to the insecticide emamectin benzoate (EB) results from the transcriptional reprogramming of a diverse suite of positive and negative regulators of juvenile hormone (JH), a critical regulator of insect development and reproduction. This in turn leads to profound increases in female BPH reproductive fitness and enhanced fecundity. Our findings unravel some of the molecular mechanisms mediating pesticide-induced pest resurgence and inform the development of novel strategies to control highly damaging crop pests.

Introduction

Chemical pesticides remain the primary means of controlling many of the world's most damaging arthropod crop pests (Janssen and van Rijn, 2021 [↗](#); Wu et al., 2018a [↗](#)). However, pesticide applications can result in pest resurgence, increases in pest insect populations that exceed natural, untreated population sizes, following an initial reduction of the pest population (Gandara et al., 2024 [↗](#); Guedes et al., 2016 [↗](#); Hardin et al., 1995 [↗](#); Wu et al., 2020 [↗](#)). Two mechanisms have been implicated in pest resurgence - the loss of beneficial insects including natural enemies, and pesticide-enhanced reproduction in the pest insect (Wu et al., 2020 [↗](#)). In the case of the latter, several pesticides, such as the insecticides triazophos, deltamethrin and the fungicide jinggangmycin, have been reported to stimulate reproduction in pest insects (Ge et al., 2010 [↗](#); Wu et al., 2018b [↗](#); Zhang et al., 2014 [↗](#)). Pesticide-enhanced pest reproduction has been linked to changes in the physiology and biochemistry of pest organisms after exposure to pesticides (Guedes et al., 2016 [↗](#); Wu et al., 2020 [↗](#)). However, the molecular mechanisms underlying enhanced reproduction associated with pest resurgence remain poorly resolved.

The brown planthopper (BPH), *Nilaparvata lugens* (Stål), is a notorious pest of rice crops throughout Asia causing annual losses of ~300 million dollars across major rice producing countries (Wu et al., 2020 [↗](#); Wu et al., 2018a [↗](#)). BPH inhibit the growth of rice plants by feeding on the phloem, and also transmit highly damaging plant viruses including rice grassy stunt virus and rice ragged stunt virus (Sōgawa, 1982 [↗](#)). Currently, chemical insecticides play an indispensable role in the control of BPH due to their efficiency, rapid effect, and low cost. However, due to the widespread and intensive use of chemical insecticides, BPH has developed resistance to the majority of compounds used for control (Wu et al., 2018a [↗](#); Zeng et al., 2023 [↗](#)).

Emamectin benzoate (EB) and abamectin are avermectin pesticides, and act as allosteric modulators of insect glutamate gated chloride channels (GluCl α) (Wolstenholme, 2012 [↗](#)), thereby inhibiting muscle contractions that lead to the cessation of insect feeding and subsequent death (Ishaaya et al., 2002 [↗](#)). These insecticides exhibit particularly strong effect against Lepidoptera such as the rice leaffolder, *Cnaphalocrocis medinalis* Guénee, an important foliage-feeding insect which attacks rice during the vegetative stage (Chintalapati et al., 2016 [↗](#)). Both BPH and the rice

leaffolder are migratory pests with overlapping migratory paths, however, their occurrence period in the field differs by approximately one month, with leaffolders appearing earlier than BPH. Therefore, the use of EB to control rice leaffolder has the potential to impact BPH arriving later, via exposure to sublethal concentrations of this compound. It has been observed that when farmers use EB and abamectin to control leaffolders on rice crops in China, BPH outbreaks frequently occur in the same field. While sublethal doses of certain pesticides have been shown to enhance fecundity in BPH, including the insecticides triazophos and deltamethrin (Ge et al., 2013 [↗](#); Ge et al., 2010 [↗](#); Zhang et al., 2014 [↗](#); Zhang et al., 2022b [↗](#)) and the fungicides carbendazim and jinggangmycin (Wu et al., 2018b [↗](#)), whether avermectins trigger resurgence in BPH via insecticide-enhanced reproduction remains unclear.

Reproduction in insects is influenced by external factors such as light (Wang et al., 2021 [↗](#)), temperature (Meiselman et al., 2022 [↗](#)), humidity (Roy et al., 2015 [↗](#)) and nutrition (Smykal and Raikhel, 2015 [↗](#)), as well as endogenous factors such as juvenile hormone (JH) (Santos et al., 2019 [↗](#)), ecdysone (Hun et al., 2022 [↗](#)), insulin- (Ling and Raikhel, 2018 [↗](#)) and target of rapamycin (TOR) signaling pathways (Ahmed et al., 2020 [↗](#); Badisco et al., 2013 [↗](#); Du et al., 2022 [↗](#); Lu et al., 2016 [↗](#)). Of these, JH has been particularly implicated in insecticide-induced enhanced fecundity, with triazophos and deltamethrin treatments leading to increased circulating JH III titers in BPH females (Wu et al., 2020 [↗](#)). JH is synthesized and secreted by the corpora allata in insects (Luo et al., 2021 [↗](#)), and promotes reproduction by regulating the biosynthesis and secretion of vitellogenin (Vg) in the female fat body, and stimulating the absorption of Vg by the developing oocytes (Santos et al., 2019 [↗](#)). However, the regulation of JH is complex (Riddiford, 2008 [↗](#); Santos et al., 2019 [↗](#)) and the key actors involved in JH-mediated pesticide-enhanced reproduction remain to be resolved. Previous findings reported that the GluCl receptor is involved in JH biosynthesis in cockroaches (Chiang et al., 2002a [↗](#); Chiang et al., 2002b [↗](#); Liu et al., 2005 [↗](#)). However, it is still unknown whether insecticides that target the GluCl α can increase reproduction in pest insects and the mechanisms by which GluCl α receptors regulate JH production remain to be unraveled.

In this study, we used a range of approaches to investigate the impact of sublethal doses of avermectins on BPH fecundity, and unravel the molecular mechanisms mediating enhanced reproduction following exposure to this insecticide class. We show that avermectin exposure results in profound changes in the expression of a key suite of genes that in combination regulate JH, which results in increased JH titer in adult females, which promotes fecundity. Interestingly, we found that emamectin benzoate (EB), known as an allosteric GluCl α channel modulators, can downregulate AstA/AstAR expression and thereby promotes JH production.

Results

Allosteric modulators (emamectin benzoate and abamectin) of the GluCl α channel stimulate fecundity of female *N. lugens*

To investigate whether GluCl α modulators affect fecundity in BPH, we first determined the sublethal and median lethal doses of emamectin benzoate (EB) on 4th instar nymphs, as well as on newly emerged males and females of BPH (Table S1). For this we employed two different bioassay methods, the rice seedling dip bioassay method and topical application bioassay method (Wu et al., 2018a [↗](#); Yao et al., 2017 [↗](#)), in order to assess both the systemic and contact toxicity of these insecticides (Table S1). We then systemically treated 4th instar nymphs of BPH, newly emerged males and females with the estimated LC₁₅ or LC₅₀ concentrations of EB and examined the fecundity of BPH after these individuals mated with treated or untreated individuals.

We use the term “t” to represent individuals treated with EB and “ck” to indicate control individuals that were treated with insecticide diluent minus insecticide. First, we tested whether treatment of BPH with EB in the nymphal stage impairs the fecundity of BPH at the adult stage and thus exposed 4th instar BPH nymphs to the pesticide. After exposure to the LC₁₅ and LC₅₀ concentrations of EB, the number of eggs laid per female of BPH in ♀t × ♂t crosses increased by 1.48- and 1.40-fold compared with control ♀ck × ♂ck crosses (**Figure 1A**); the number of eggs laid per female of BPH in ♀t × ♂ck crosses increased by 1.53- and 2.07- fold compared with control crosses (**Figure 1B**); However, the number of eggs laid by per female of BPH in ♀ck × ♂t crosses did not increase significantly compared to control ♀ck × ♂ck crosses (**Figure 1C**).

Exposure of 4th instar nymphs to the LC₁₅ and LC₅₀ concentrations of EB in contact bioassays also significantly stimulated fecundity. After treatment with the LC₁₅ and LC₅₀ concentrations of EB, the number of eggs laid by per female of BPH in ♀t × ♂t crosses increased by 1.18- and 1.26-fold compared with the control (♀ck × ♂ck) (**Figure 1D**); The number of eggs laid per female of BPH in ♀t × ♂ck crosses increased by 1.27- and 1.56-fold compared with the control crosses (**Figure 1E**); However, there was no significant difference in number of eggs laid between ♀ck × ♂t crosses and controls (♀ck × ♂ck) (**Figure 1F**). These results reveal that EB stimulates the fecundity of females following both systemic and contact routes of exposure.

Next, we examined whether EB treatment of adult BPH also enhances reproduction in relation to egg laying. Indeed, treating newly emerged adults with the LD₁₅ and LD₅₀ concentrations of EB significantly stimulated the number of eggs laid per female (**Figure 1-figure supplement 1A**). Furthermore, sub-lethal exposure of 4th instar BPH nymphs to another allosteric modulator of GluCl α , abamectin (LC₁₅ and LC₅₀ concentrations), was also found to significantly enhance oviposition (**Figure 1-figure supplement 1B**).

To determine whether EB administered during the fourth-instar larval stage persists as residues in the adult stage, we used HPLC-MS/MS to quantify the amount of EB present at the adult stage after exposing 4th-instar nymphs to this compound.

However, we found no detectable EB residues in the adult stage following fourth-instar nymphal treatment (**Figure 1-figure supplement 1C**). This suggests that the mechanism underlying the increased fecundity of female adults induced by EB treatment of nymphs may differ from that caused by direct EB treatment of female adults. Combined with our previous observation that EB treatment significantly increased the body weight of adult females (**Figure 1—figure supplement 3E and F**), a possible explanation for this phenomenon is that EB may enhance food intake in BPH, potentially leading to elevated production of insulin-like peptides and thus increased growth. Increased insulin signaling could potentially also stimulate juvenile hormone (JH) biosynthesis during the adult stage (Badisco et al., 2013).

To examine if EB also stimulates egg-laying in other insect species we conducted bioassays on the small brown planthopper, *Laodelphax striatellus*, the white backed planthopper, *Sogatella furcifera* and vinegar flies *Drosophila melanogaster*. In contrast to our findings in BPH, we found that sub-lethal doses (LC₁₅ and LC₅₀) of EB inhibits fecundity of female *L. striatellus*, (**Figure1-figure supplement 2A-C**) and has no impact on the fecundity of *S. furcifera*, (**Figure1-figure supplement 2D-F**). In addition, we found that sublethal doses (LC₁₅, LC₃₀ or LC₅₀) of EB also inhibit fecundity in *D. melanogaster* (**Figure1-figure supplement 2G and H**). These results indicate that the stimulation of reproduction by EB in BPH is species-specific and does not extend to even related insect species.

The impact of EB treatment on BPH reproductive fitness

To better understand the effects of EB on the reproductive fitness of BPH, the duration of the preoviposition period, emergence rate, female ratio, female longevity and female weight were evaluated following exposure using systemic bioassays. The preoviposition period of females

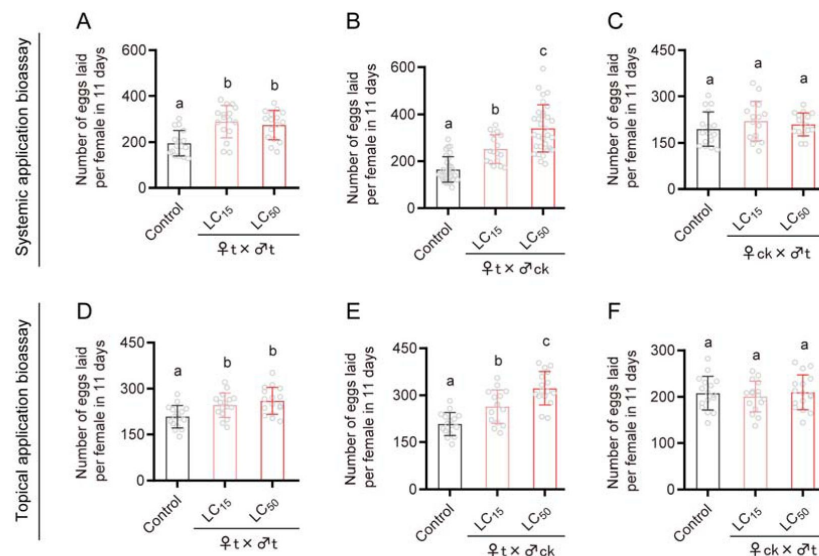


Figure 1.

Fecundity of BPH following exposure to sub-lethal (LC₁₅) and median lethal (LC₅₀) concentrations of emamectin benzoate following systemic application bioassays (A: ♀ t × ♂ t; B: ♀ t × ♂ ck; C: ♀ ck × ♂ t) and topical application bioassays (D: ♀ t × ♂ t; E: ♀ t × ♂ ck; F: ♀ ck × ♂ t), respectively.

The letter “t” represents treatment with insecticide, while “ck” indicates controls that were not treated with insecticide. Insects were exposed as 4th instar nymphs. All data are presented as the mean ± s.e.m. Different lower-case letters above the bars indicate significant differences (One-way ANOVA with Tukey’s Multiple Range Test, $p < 0.05$).

treated with the LC_{50} of EB decreased significantly compared with the control (**Figure 1-figure supplement 3A** [↗](#)). In contrast no significant effects of EB on emergence rate and female ratio were observed (**Figure 1-figure supplement 3B and C** [↗](#)). Female survival analysis showed that exposure of 4th instar nymphs to the LC_{50} of EB has no impact on female longevity (**Figure 1-figure supplement 3D** [↗](#)). Interestingly, occurrence of brachypterism (short-wing phenotype) in females and female weight were significantly increased after EB exposure (**Figure 1-figure supplement 3E and F** [↗](#)). It was reported that short-winged females usually display higher fecundity than long-winged females in BPH and two different insulin receptors determine the alternative wing morphs (Xu et al., 2015 [↗](#)). However, whether insulin signaling is involved in the fecundity increase of BPH in our experiments requires further investigation (Ling and Raikhel, 2021 [↗](#); Mirth et al., 2014 [↗](#); Sheng et al., 2011 [↗](#)).

EB promotes ovarian maturation in BPH

To investigate the cause of increased egg-laying following EB exposure, we examined whether EB influences ovarian maturation in BPH. We dissected and compared the ovaries of females treated with the LC_{50} of EB at 1, 3, 5, and 7 days after eclosion (DAE) with control females. At 3, 5, and 7 DAE, the number of eggs in the ovary of BPH in the EB treated group was significantly higher than in controls (**Figure 2A and B** [↗](#)). We furthermore observed that EB treatment of female adults also increases the number of mature eggs in the ovary (**Figure 2-figure supplement 1** [↗](#)).

Hence, although EB does not affect the normal egg developmental stages (see description in next section), our results suggest that EB treatment promotes oogenesis and, as a result the insects both produce more eggs in the ovary and a larger number of eggs are laid.

Unlike the model insect *Drosophila melanogaster*, studies on oogenesis in BPH are relatively limited. Therefore, we used immunohistochemistry to stain eggs at different stages of BPH development. We captured images with a laser confocal microscope to document each stage of oogenesis in this species. We found that oogenesis in the BPH differs from that in *D. melanogaster* and belongs to the telotrophic meroistic type, where the oocyte is connected to nutrient-providing cells in the anterior trophic region of the ovariole through a nutritive cord. Oogenesis in the BPH occurs in three main compartments (**Figure 2-figure supplement 2** [↗](#)):

1. Terminal Filament (TF): This region, composed of terminal filament cells, is similar to the terminal filament structure found in *Drosophila*.
2. Germarium: This region is where germline stem cells (GSCs) differentiate into oocytes and follicle cells. It typically consists of cap cells adjacent to the terminal filament, germline stem cells, escort cells, and cystoblasts.
3. Egg Chamber: The egg chamber in the BPH primarily contains eggs at six developmental stages. The distinct layering of follicle cells, visible through cytoskeletal staining, is the main feature distinguishing the six different developmental stages listed in the legend of **Figure 2-figure supplement 2** [↗](#).

We next explored whether EB treatment can enhance or impair oogenesis in BPH. However, dissection of various developmental stages revealed that EB treatment has no significant effects on developmental stages and morphology of oogenesis in BPH (**Figure 2C and D** [↗](#)). However, considering that the number of eggs laid by EB treated females was larger than in control females (**Figure 1** [↗](#) and **Figure 1-figure supplement 1** [↗](#)), our data indicates that EB treatment of BPH can both promote both oogenesis and oviposition.

Stage 1: Connected to the germarium, the egg is trapezoidal. Stage 2: The egg is larger than in Stage 1 and is square to rectangular. Stage 3: The egg begins to elongate toward both poles, taking on an oval shape. Stage 4: The egg continues to enlarge and elongate, becoming spindle-shaped. Due to rapid growth, follicle cells on the egg surface start dividing, indicating nuclear division. Stage 5:

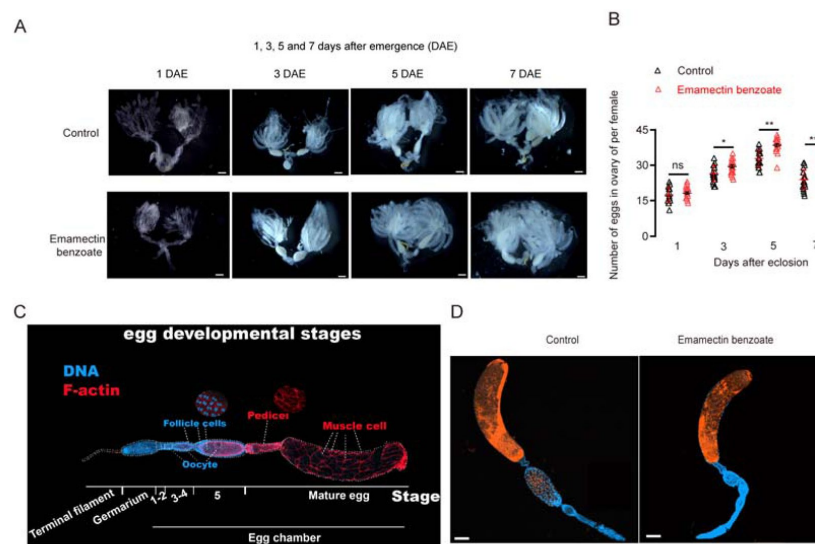


Figure 2.

The impact of emamectin benzoate on ovarian maturation in BPH.

Fourth instar nymphs were treated with the LC_{50} concentration of emamectin benzoate in systemic bioassays. (A) Ovarian development in EB treated BPH at 1, 3, 5 and 7 days after eclosion (DAE) compared to untreated controls. Scale bar = 1,000 μ m. (B) Number of mature eggs in the ovaries of EB treated 4th instar BPH nymphs measured at 1, 3, 5 and 7 DAE compared to controls. All data are presented as the mean $\pm$ s.e.m. Asterisks indicate values significantly different from the control using student *t* test (ns, no significant; **p* < 0.05 and ***p* < 0.01). (C) Different developmental stages of BPH eggs. (D) No impairment of emamectin benzoate on oogenesis of BPH. Scale bar = 100 μ m.

The egg further enlarges, and most follicle cells display binucleation, the most notable feature of this stage. Small amounts of lipids start entering the egg. Stage 5 is the final stage before maturation, after which the egg enters the pedicel, a structure primarily formed by muscle fibers. The expanded mature egg stretches the muscle fibers in the pedicel, increasing the spacing between them. Stage 6: This is the mature egg stage, which occurs after the egg passes through the pedicel. During this stage, the egg undergoes rapid growth, continuously absorbing vitellogenin and lipids (Nile red staining confirmed that Stage 6 has the highest lipid content). This absorption causes the egg's surface color to transition from the transparency seen in Stage 5 to mature eggs' dark, opaque appearance. Once the egg enters the pedicel and becomes completely opaque, it reaches full maturity. The mature egg then moves into the lateral oviduct for oviposition, completing the egg-laying process.

EB exposure increases circulating sugars and the storage of macromolecules in BPH

Nutritional status is an important indicator of reproductive fitness. Thus, to investigate whether EB affects intermediary metabolism and energy storage in BPH, glycogen, triacylglyceride (TAG) total protein content, cholesterol and four circulating carbohydrates were quantified in 4th instar BPH nymphs following exposure to the LC₅₀ of EB.

We found that EB exposure has no impact on glycogen levels (**Figure 2-figure supplement 2A** [↗](#)). The amount of TAG in EB-treated BPH was 27% higher ($p < 0.05$) than those in controls, but only in BPH of the late fifth instar (5L) stage, with no significant differences observed in subsequent developmental stages (**Figure 2-figure supplement 2B** [↗](#)). The amount of total protein content in EB-treated BPH was higher than the control groups in the case of all developmental stages from 5L nymph to 7DAE (**Figure 2-figure supplement 2C** [↗](#)). EB exposure also increased cholesterol levels at 4 and 5 DAE (**Figure 2-figure supplement 2D** [↗](#)). Compared with the solvent control, EB treatment caused significant increases ($p < 0.05$) in the levels of sucrose, glucose, fructose, and trehalose (**Figure 2-figure supplement 2E-H** [↗](#)). Thus, collectively, these data provide evidence that EB exposure leads to energy mobilization such as an increase in circulating lipids and carbohydrates in BPH.

EB stimulates egg-laying that is mediated by the JH signaling pathway

Given the important role of juvenile hormone (JH) in vitellogenesis and egg development in insects ([Jing et al., 2021](#) [↗](#); [Ling and Raikhel, 2021](#) [↗](#); [Luo et al., 2021](#) [↗](#); [Santos et al., 2019](#) [↗](#); [Zheng et al., 2022](#) [↗](#)), we asked whether EB-treatment influences the titer of JH in BPH. As measured by ELISA, the juvenile hormone titer of BPH nymphs treated with the LC₅₀ concentration of EB was significantly lower than that of controls in systemic bioassays during the middle and late stages of the 4th instar (**Figure 3A** [↗](#)). However, at 2, 3 and 4 DAE, the JH titer in the EB treated group was significantly higher than that of the control (**Figure 3A** [↗](#)). Interestingly, the titer of another important insect hormone, the steroid 20-hydroxyecdysone, was not significantly different between EB-treated BPH and solvent-treated controls (**Figure 3-figure supplement 1** [↗](#)). To independently validate the results of the ELISA assays, we employed HPLC-MS/MS to measure the JH titer in BPH following EB exposure ([Bownes and Rembold, 1987](#) [↗](#); [Guo et al., 2020](#) [↗](#); [Jing et al., 2021](#) [↗](#)). The results showed that the JH III titer significantly decreased after EB-treatment at the late 4th instar nymph stage (**Figure 3A and B** [↗](#)), but significantly increased at the third day after eclosion (3 DAE) (**Figure 3A and C** [↗](#)). We also investigated the effects of EB treatment on the JH titer of female adults. The data indicate that the JH titer was also significantly increased in the EB-treated female adults compared with controls (**Figure 3-figure supplement 3A** [↗](#)). However, again the steroid 20-hydroxyecdysone, was not significantly different between EB-treated BPH and controls (**Figure 3-figure supplement 3B** [↗](#)).

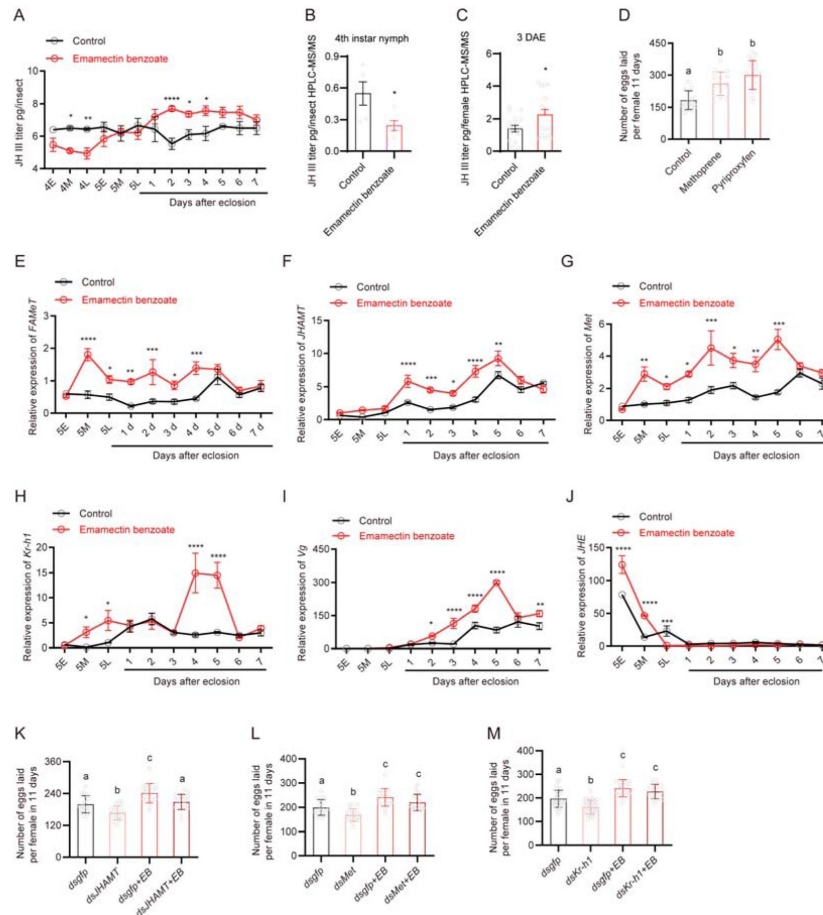


Figure 3.

EB induced reproduction in BPH is mediated by components of the JH signaling pathway.

(A) The titer of JH III (as measured by ELISA assay) at different developmental stages of whole body BPH when 4th instar nymphs were treated with the median lethal concentrations of EB. (B and C) The titer of JH III (as measured by HPLC-MS/MS) in whole body BPH females at 4L and 3 DAE when treated with median lethal concentrations of EB. (D) Oviposition rate of BPH when 4th instar nymphs were treated with 4 ppm methoprene or 10 ppm pyriproxyfen. (E-J) Expression of selected JH-related genes (*FAMeT*, *JHAMT*, *Met*, *Kr-h1*, *Vg*, and *JHE*) in EB-treated BPH. (K) Egg production following silencing of *JHAMT* with or without EB application. (L) Egg production following silencing of *met* with or without EB application. (M) Egg production after silencing *Kr-h1* with or without EB application. All data are presented as means $\pm$ s.e.m. Student's *t* test was used to compare controls and treatments. One-way ANOVA with Tukey's multiple comparisons test was used to compare more than two samples. ns, no significant difference; Asterisks indicate values significantly different from the control (ns, no significant; **p* < 0.05, ***p* < 0.01, ****p* < 0.001 and *****p* < 0.0001). Different lower-case letters above the bars indicate significant differences (*p* < 0.05).

To further investigate the role of JH in EB-enhanced fecundity in BPH, we treated BPH with methoprene and pyriproxyfen, JH analogues or biologically active JH mimic, respectively, to determine whether they can stimulate fecundity. Both compounds significantly increased egg-laying in BPH (**Figure 3D**). Taken together, these results reveal that EB exposure is associated with an increase in JH titer and that this elevated JH signaling contributes to enhanced fecundity in BPH.

Since we found that EB could induce JH synthesis in the BPH, we asked whether EB could influence the expression of genes that are involved in JH biosynthesis or degradation. For this we treated 4th instar nymphs or newly emerged female adults of BPH with the LC₅₀ concentration of EB using systemic bioassay, and then collected early (5E), middle (5M) and late (5L) stage of 5th instar nymph and 1-7 DAE female adults for analysis. Quantitative PCR was then used to examine the expression of key genes previously implicated in the regulation of JH (Gospocic et al., 2021; Nouzova et al., 2018).

Farnesoic acid O-methyltransferase (FAMeT) (Liu et al., 2010b), is known as an important enzyme in the JH biosynthetic pathway, catalyzing methylation of farnesoic acid (FA) to methyl farnesoate (MF) (Nouzova et al., 2018). We found that this gene was significantly upregulated in 5M instar nymphs to 4 DAE following EB treatment (1.0-fold to 3.0-fold) (**Figure 3E**). Juvenile hormone acid methyltransferase (JHAMT) (Nouzova et al., 2018; Shinoda and Itoyama, 2003), which is involved in the biosynthesis of JH, was also upregulated in EB-treated BPH at 1 to 5 DAE compared to controls (1.5-fold to 3.0-fold) (**Figure 3F**). We furthermore found that EB treatment in female adults increases JHAMT expression (**Figure 3-figure supplement 3C**).

Methoprene-tolerant (*Met*), belongs to the basic helix–loop–helix Per/Arnt/Sim (bHLH-PAS) family of transcription factors and is the intracellular (nuclear) receptor of JH (Riddiford, 2020; Zhu et al., 2019). The levels of *met* mRNA slightly increased in EB-treated BPH at the 5M and 5L instar nymph and 1 to 5 DAE adult stages compared to controls (1.7-fold to 2.9-fold) (**Figure 3G**). However, it should be mentioned that JH action does not result in an increase of *Met*. Thus, it is possible that other factors (indirect effects), induced by EB treatment cause the increase in the mRNA expression level of *Met*.

Krüppel homolog 1 (*Kr-h1*), a transcriptional target of JH signaling, is reported to be sensitive to levels of JH and its expression levels are directly correlated with JH titers (Gospocic et al., 2021; Jindra et al., 2013; Zhang et al., 2022a). We found that *Kr-h1* was significantly upregulated in the adults of EB-treated BPH at the 5M, 5L nymph and 4 to 5 DAE stages (4.7-fold to 27.2-fold) when 4th instar nymph or female adults were treated with EB (**Figure 3H** and **Figure 3-figure supplement 3D**).

Similarly, the expression level of *vitellogenin* (*Vg*), a key downstream component of JH signaling that triggers ovarian development in insects, including BPH (Shen et al., 2019), was markedly increased in females at 2–5 DAE by EB (1.7-fold to 5.5-fold) (**Figure 3I**). During the maturation process, eggs require uptake of *Vg*, and an increase in *Vg* content can accelerate egg maturation, producing more mature eggs. Since our molecular data suggest that EB treatment leads to an upregulation of *Vg* expression, we propose that the increase in egg-laying caused by EB treatment is at least in part due to the upregulation of *Vg* (**Figure 3I**), which elevates *Vg* content, leading to increased uptake of *Vg* by maturing eggs and resulting in the production of more mature eggs.

Juvenile hormone esterase (JHE) is the primary JH-specific degradation enzyme that plays a key role in regulating JH titers (Kamita et al., 2003). Interestingly, we observed a significant upregulation of JHE mRNA levels in the early and middle 5th instar nymph stage, followed by downregulation in 5L instar nymphs of EB-treated BPH (1.3-fold to 2.6-fold) (**Figure 3J**).

Taken together, these results suggest that EB has a profound impact on the expression of key genes involved in the biosynthesis of JH or genes in the downstream signaling pathway genes that might promote egg development and female fecundity. However, our data do not allow us to conclude whether the effect of EB on the genes regulating the JH titer is direct or indirect.

To further understand whether these JH pathway-related genes are involved in egg-laying behavior in BPH, we performed RNAi experiments to downregulate the expression of *Kr-h1* (Figure 3-figure supplement 2A). We found that silencing of *Kr-h1* diminished *vg* gene expression (Figure 3-figure supplement 2B). Importantly, RNAi-induced diminishment of *JHAMT*, *Met* and *Kr-h1* genes in female BPH was also found to suppress egg-laying (Figure 3K-M). However, this phenotype was rescued by EB treatment, which restored egg-laying to normal levels in BPH injected with *JHAMT*, *Met* and *Kr-h1* dsRNA (Figure 3K-M). Note that this rescue is possible since knockdown of the genes is incomplete when using dsRNA injection (and residual gene expression allows for EB action). Together these results provide a mechanistic understanding of how EB enhances fecundity in BPH by modulating the expression of key genes involved in JH synthesis.

Next, we investigated whether EB treatment could rescue the dsRNA-mediated gene silencing effect. To address this, we selected the *Kr-h1* gene and analyzed its expression levels after EB treatment. Our results showed that *Kr-h1* expression was suppressed by ~70% at 72 h post-dsRNA injection. However, EB treatment did not significantly rescue *Kr-h1* expression in gene knock down insects (*p* > 0.05) at 24h post-EB treatment (Figure 3-figure supplement 4). While dsRNA-mediated *Kr-h1* suppression was robust initially, its efficacy may decline during prolonged experiments. This aligns with reports in BPH, where effects of RNAi gradually diminish beyond 7 days post-injection (Liu et al., 2010a). The late-phase fecundity increase might reflect partial *Kr-h1* recovery due to RNAi degradation, allowing residual EB to weakly stimulate reproduction. In addition, the physiological impact of EB (e.g., neurotoxicity, hormonal modulation) could manifest via compensatory feedback loops or metabolic remodeling.

EB induces JH biosynthesis through the peptidergic AstA/AstAR signaling pathway

In many insect species, the timing and level of JH biosynthesis can be precisely regulated by neuropeptides, such as stimulatory allatotropins (ATs) and inhibitory allatostatins (Asts), (Bellés et al., 1999; Kataoka et al., 1989; Kramer et al., 1991; Lorenz et al., 1995; Stay and Tobe, 2007; Verlinden et al., 2015; Wegener and Chen, 2022; Woodhead et al., 1989). Insects can, in a species-specific manner, produce one type of AT and three types of Asts: FGL-amide Ast (AstA) (Woodhead et al., 1989; Yin et al., 2006); myoinhibitory peptide (MIP or AstB) (Stay and Tobe, 2007) and PISCF Ast (AstC) (Wang et al., 2012). In some species, there also exists two paralogous genes of AstCs which are named *AstCC* and *AstCCC* (Veenstra, 2009, 2016). Interestingly, the allatostatic activity of these three types of Ast peptides varies between insect species so that in each species only one type of Ast (for example AstA) controls JH production (Nässel, 2002; Stay and Tobe, 2007; Wang et al., 2012; Wegener and Chen, 2022).

Analysis of neural transcriptome sequence data of BPH has revealed the presence of one AT, four types of Asts and four corresponding receptors, allatotropin receptor (A16, ATR), AstA receptor (A2, AstAR), AstB (MIP) receptor (A10, AstBR or MIPR) and AstC receptor (A1, AstCR) (Tanaka et al., 2014). We cloned the five neuropeptide genes (encoding AT, AstA, AstB/MIP, AstCC and AstCCC) and confirmed the sequences obtained from transcriptome data (Figure 4-figure supplement 1) (Tanaka et al., 2014). Interestingly, we found that AstC is missing in the genome of BPH and only AstCC and AstCCC are present (Figure 4-figure supplement 1). Next, we also cloned their corresponding receptors (Veenstra, 2009) including ATR (A16), AstAR (A2), AstBR (A10) and

AstCR (A1) which might be activated by AstCC and/or AstCCC (Audsley et al., 2013 [DOI](#); Veenstra, 2009 [DOI](#); Zhang et al., 2022a [DOI](#)). Sequence alignments and phylogenetic analysis are shown in **Figure 4-figure supplement 2** [DOI](#).

Quantitative PCR was then used to examine whether EB treatment influences the expression of the genes encoding these neuropeptides and their receptors. Treating 4th instar or female adult BPH with the LC₅₀ concentration of EB significantly increased the expression of *AT*, *ATR* and *AstCCC*, while resulting in the downregulation of *AstA*, *AstB/mip*, *AstCC*, *AstAR* and *AstBR/mipr* at the adult stage (**Figure 4A and B** [DOI](#), **Figure 4-figure supplement 3A-G** [DOI](#)). Among these, *AstA* and *AstAR* were the most downregulated genes after EB treatment (**Figure 4A and B** [DOI](#), **Figure 4-figure supplement 3H** [DOI](#)) and thus the *AstA/AstAR* signaling system was selected for subsequent functional analysis. Silencing of *AstAR* in female BPH using RNAi (**Figure 4C** [DOI](#)), resulted in an increased number of eggs laid per female compared with *dsgfp*-injected controls (**Figure 4D** [DOI](#)). Interestingly, silencing *AstAR* also resulted in the upregulation of *JHAMT*, *Met* and *Kr-h1*, which are involved in the JH biosynthesis/signaling (**Figure 4E-H** [DOI](#)). However, *JHE* was not influenced by *AstAR* silencing (**Figure 4I** [DOI](#)). We next investigated whether silencing the *AstAR* gene influences the JH titer in BPH. Indeed, a significantly increased JH titer was observed in *AstAR* silenced BPH compared with controls (**Figure 4J** [DOI](#)). Thus, our data strongly suggest that *AstA* is a key inhibitor of JH production in BPH.

Finally, we investigated whether injection of mature *Ast* and *AT* peptides could influence the number of eggs laid and the JH titer in BPH. We synthesized one *AT*, six *AstAs* (*AstA1* to *AstA6*), one *AstCC* and one *AstCCC* peptide according to their determined sequences (**Figure 4-figure supplement 1** [DOI](#)). Indeed, we found that injection of *AstA1* and *AstA6* in female adults reduced the number of eggs laid per female over 48 h (**Figure 4K** [DOI](#)). We also observed that *AstA1* injection decreased the JH titer as measured 16 h and 48 h after injection (**Figure 4L** [DOI](#)). *AT* injection increased the JH titer after 2 h, but levels returned to normal 4 h after injection (**Figure 4M** [DOI](#)). Collectively, our data provide compelling evidence that EB enhances reproduction in BPH probably indirectly through the *AstA/AstAR* and JH signaling pathways and further supports the role of *AstA* and *AT* in regulation of the JH titer in this species.

EB-enhanced fecundity in BPH is dependent on its molecular target protein, the GluCl_α channel

EB and abamectin, are allosteric modulators, which target glutamate-gated chloride channels (GluCl_s) (Sparks et al., 2020 [DOI](#); Sparks et al., 2021 [DOI](#); Wolstenholme, 2012 [DOI](#); Wu et al., 2017 [DOI](#)). This channel is found in invertebrates only, and appears to be monomeric and formed by GluCl_α subunits (Wolstenholme, 2012 [DOI](#)). Hence, we examined whether EB-stimulated fecundity in BPH is mediated by its molecular target GluCl_α. The full length *GluCl_α* coding sequence from BPH was cloned and sequenced (**Figure 5-figure supplement 1** [DOI](#)) and the impact of EB on *GluCl_α* gene expression examined using quantitative PCR. Treatment of BPH with the LC₅₀ concentration of EB significantly downregulated *GluCl_α* gene expression at the 5E and 5M nymph stages, while upregulating *GluCl_α* gene expression at 2 DAE and 5 DAE in the adult stage (**Figure 5A** [DOI](#)). We also found that EB treatment in female adults also increases the *GluCl_α* gene expression (**Figure 5-figure supplement 3** [DOI](#)). However, we do not propose that EB is a direct transcriptional regulator of *GluCl_α*, since EB and other avermectins are known to alter the channel conformation and thus their function (Wolstenholme, 2012 [DOI](#); Wu et al., 2017 [DOI](#)). Thus, it is likely that the observed increase in *GluCl_α* transcript is a secondary effect downstream of EB signaling.

To examine the role of *GluCl_α* gene in BPH fecundity, RNAi was used to knock down expression of this gene in female adult BPH (**Figure 5B** [DOI](#)). A significant decrease in the number of eggs laid by per female was observed in *dsGluCl_α*-injected insects compared with *dsgfp*-injected insects (**Figure 5C** [DOI](#)). However, treatment with EB was found to partially rescue the decreased egg-laying phenotype induced by *dsGluCl_α* injection (**Figure 5C** [DOI](#)). Note that this rescue is possible since

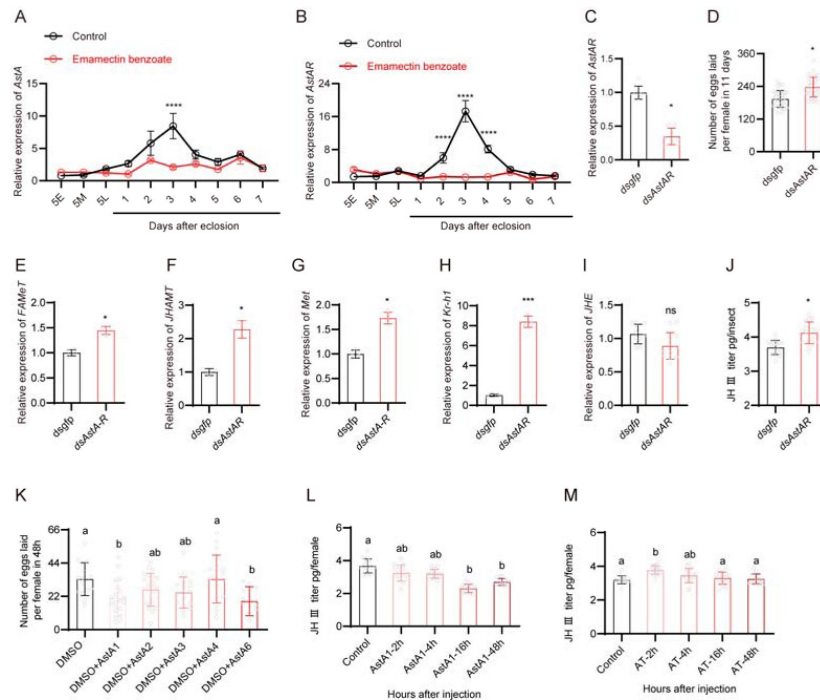


Figure 4.

EB induced reproduction in BPH is mediated by the *AstA/AstAR* and JH signaling pathway.

(A and B) Expression of *AstA* and *AstAR* in different stages of BPH following EB treatment. (C) Downregulation of *AstAR* using RNAi leads to a reduction in mRNA expression level. (D) Egg production in female BPH following silencing of *AstAR* gene. (E-I) Expression of selected JH signaling pathway related genes (*JHAMT*, *Met*, *Kr-h1* and *JHE*) in *AstAR* silenced BPH. (J) JHIII titer of BPH females after *AstAR* gene silencing determined by HPLC-MS/MS. (K) Number of eggs laid per female in 48h following injection of the six mature *AstA1-AstA6* peptides and one mature AT peptide. Fifty nanoliter of PBS (as control) and seven different peptides (20 pmol/insect) were injected into female BPH three days after eclosion. (L and M) The JH III titer of whole body BPH females at different time points following *AstA* or AT injection. All data are presented as means $\pm$ s.e.m. Student's *t* test was used to compare controls and treatments. One-way ANOVA with Tukey's multiple comparisons test was used to compare more than two samples. ns, no significant difference; Asterisks indicate values significantly different from the control (ns, no significant; **p* < 0.05, ***p* < 0.01, ****p* < 0.001 and *****p* < 0.0001). Different lower-case letters above the bars indicate significant differences (*p* < 0.05).

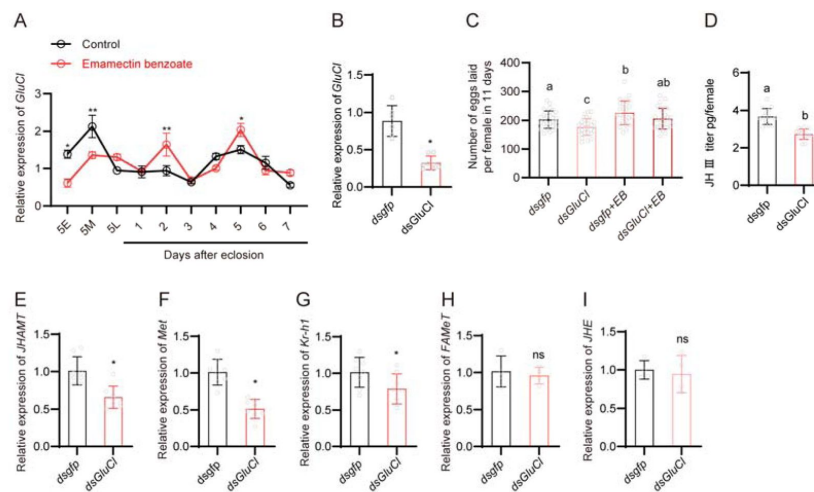


Figure 5.

Role of EB and the *GluCl* channel in fecundity and juvenile hormone signaling in BPH

(A) Expression of *GluCl* in EB-treated and untreated BPH. (B) Expression of *GluCl* following injection of ds*GluCl* in BPH. (C) Egg production after *GluCl* gene knockdown in EB-treated and untreated BPH. (D) The JH III titer of whole body BPH females after *GluCl* gene silencing as quantified using the ELISA method. (E-I) Expression patterns of selected JH-related genes (*JHAMT*, *Met*, *Kr-h1* and *JHE*) in *GluCl* silenced BPH. All data are presented as means $\pm$ s.e.m. Student's t test was used to compare the two samples. One-way ANOVA with Tukey's multiple comparisons test was used to compare more than two samples. ns, no significant difference; Asterisks indicate values significantly different from the control (ns, no significant; * $p < 0.05$ and ** $p < 0.01$). Different lower-case letters above the bars indicate significant differences ($p < 0.05$).

knockdown of the genes is incomplete when using dsRNA injection (and residual gene expression allows for EB action). To investigate the mechanism by which *GlucI_α* expression affects fecundity we examined if silencing *GlucI_α* influences JH titer and JH-associated gene expression. Indeed, we observed that RNAi knockdown of *GlucI_α* leads to a decrease in JH titer (**Figure 5D**) and down-regulation of genes including *JHAMT* which is responsible for JH synthesis, and the JH signaling downstream genes *Met* and *Kr-h1* (**Figure 5E-G**). In contrast, expression of *JHE* was not changed in the *GlucI_α* knockdown insects (**Figure 5H and I**). We also examined whether silencing *GlucI_α* impacts the *AstA/AstAR* signaling pathway in female adults. Knock-down of *GlucI_α* in female adults was found to have no impact on the expression of *AT*, *AstA*, *AstB*, *AstCC*, *AstAR*, and *AstBR*. However, the expression of *AstCCC* and *AstCR* was significantly upregulated in *dsGlucI_α*-injected insects (**Figure 5-figure supplement 2A-H**). Further studies are required to delineate the direct or indirect mechanisms underlying this effect of *GlucI_α*-knockdown.

Discussion

Pesticide-induced resurgence of pest insects is a serious problem in rice and several other crops (Wu et al., 2020). However, the mechanisms underpinning pesticide-enhanced reproduction in insects remain poorly understood. Here we reveal that a suite of molecular actors underlie this trait that, in combination, mediate profound physiological changes in the reproductive fitness of female BPH. Our data provide fundamental insights into the molecular mechanisms by which xenobiotics modify insect reproduction and have implications for devising strategies for the control of a highly damaging crop pest. We discuss these topics below.

Sublethal doses of the GluCl modulators, EB and abamectin, stimulates fecundity in BPH

We show that in both contact and systemic assays the avermectins EB and abamectin stimulate reproduction in BPH. Thus, insecticide-enhanced reproduction is likely a key factor in the BPH resurgence observed when farmers use EB and abamectin to control leafhoppers in rice crops in China (Yao et al., 2017). Although this is the first report of sublethal doses of avermectins enhancing insect fecundity, our findings are consistent with previous studies, which have shown that certain other insecticides, herbicides and fungicides stimulate BPH reproduction (Cheng et al., 2014; Ge et al., 2011; Ge et al., 2013; Wan et al., 2013; Wang et al., 2010; Wu et al., 2020; Yang et al., 2019; Zhang et al., 2014; Zhao et al., 2011). Intriguingly, we show that EB only induces fecundity in female adults and is specific to BPH, with EB exposure failing to enhance reproduction in two related species, the small brown planthopper, *L. striatellus* and the white backed planthopper, *S. furcifera*, or the model insect *D. melanogaster*. Thus, the mechanisms underpinning this trait appear to act exclusively on female BPH and may be specific to this species. Nevertheless, our findings are of general interest since they link the molecular EB target, the GluCl channel and juvenile hormone signaling to fecundity. Furthermore, our preliminary experiments suggest that EB diminishes fecundity in *Drosophila* (see **Figure 1 – Supplement 2G, H**), a finding worth pursuing in this genetically tractable insect. It can be noted that the GluCl channels were first identified and cloned in the nematode worm *Caenorhabditis elegans*, where they were also found to be targets of avermectins and involved in reproduction and fecundity (Wolstenholme, 2012).

Pesticides may stimulate insect reproduction through a variety of physiological and molecular mechanisms. Our data reveal that exposure to sub-lethal concentrations of EB results in profound changes to female BPH fitness, leading to increases in female weight, contents of total protein, cholesterol, and sugars, as well as egg production and decreases in duration of the preoviposition period. Some of these findings exhibit parallels with previous studies, which demonstrated that treating third-instar BPH nymphs with either deltamethrin, triazophos, or imidacloprid led to increased soluble sugar levels in the corresponding adults (Yin et al., 2014). Such metabolites

provide the energy that drives BPH reproduction and hence resurgence (Wu et al., 2020 [↗](#)). Thus, together with prior work, our results suggest that pesticides associated with resurgence stimulate nutritional intake in BPH to fuel enhanced energy-intensive reproduction.

The JH signaling pathway plays an essential role in EB-induced fecundity in BPH

JH is a pleiotropic hormone, which plays important roles in development and reproduction in insects (Jindra et al., 2013 [↗](#); Riddiford, 2008 [↗](#); Santos et al., 2019 [↗](#)). Circulating JH titers are regulated by factors that control JH production in the corpora allata including biosynthetic enzymes and catabolic enzymes that regulate JH levels. Our results show that EB exposure is associated with increased circulating JH III titers in BPH females over 2–4 days after eclosion (DAE) and promotes ovarian development. Previous studies have reported that triazophos and deltamethrin treatments also lead to increased circulating JH III titers in BPH females over 1–3 days post emergence. Similarly, jinggangmycin treatments were found to lead to increased JH titers (by approximately 45–50%) in BPH females over two days post emergence (Xu et al., 2016 [↗](#)). Thus, our findings, in combination with these previous studies, demonstrate that insecticide treatments can have dramatic effects on the regulation of key insect hormones involved in pest reproduction, which can in turn drive pesticide resurgence.

Although increased JH titers following pesticide exposure have been correlated with reduced levels of active JH esterase during the first three days PE (Ge et al., 2010 [↗](#)), the types and numbers of mechanisms mediating the observed increase in hormone titer has remained an open question. Our data reveal that the elevated JH titer in EB-exposed BPH is associated with the upregulation a gene that encode a biosynthetic enzyme for JH (*JHAMT*) and a downstream signaling gene that can induce *vg* gene expression (*kr-h1*). Using RNAi we provide experimental evidence of the role of these genes in the regulation of JH III and fecundity of female BPH, and demonstrate that EB can restore the reduction in egg production resulting from the partial RNAi knockdown of *JHAMT*, *met* and *kr-h1* expression.

JHAMT is an enzyme that catalyzes the conversion of inactive precursors of JH to active JH in the final stages of JH biosynthesis (Nouzova et al., 2018 [↗](#); Shinoda and Itoyama, 2003 [↗](#)). Interestingly, while it has not been previously implicated in pesticide resurgence, treatment of the stored product pest *Sitotroga cerealella* with diallyl trisulfide, an insecticidal compound in garlic essential oil, was found to increase JHAMT mRNA levels (Shah et al., 2022 [↗](#)). Because JHAMT is the key rate-limiting enzyme in regulation of JH titer our results suggest that its enhanced expression is a key molecular mechanism of pesticide resurgence in BPH.

Met is a ligand-activated member of the basic helix–loop–helix Per/Arnt/Sim (bHLH-PAS) transcription factors and is the intracellular receptor for JH (Charles et al., 2011 [↗](#); Jindra et al., 2015 [↗](#); Riddiford, 2020 [↗](#); Zhu et al., 2019 [↗](#)). *Kr-h1* is a zinc finger protein that acts downstream of Met and is expressed in response to JH signaling. Although the genes encoding these proteins have not been previously linked to pesticide resurgence, our finding that they are upregulated following EB exposure, and demonstration of their role of in promoting fecundity, is consistent with previous studies (Santos et al., 2019 [↗](#)). Specifically, treatment of BPH with JH III or the insecticidal analogs methoprene or pyriproxifen was found to induce the expression of *Kr-h1* (Jin et al., 2014 [↗](#)). Furthermore, knockdown of *Met* and *Kr-h1* in brachypterous BPH females was found to result in delayed ovariole maturation and this was significantly more pronounced than the response observed in BPH treated separately with *dsNlMet* or *dsNlKr-h1* (Lin et al., 2015 [↗](#)). This finding provides evidence of a possible interaction between *Met* and *Kr-h1* and, in combination with our data, suggests that *Met* and *Kr-h1* may act in concert to mediate EB-enhanced fecundity. However, our results showed that EB treatment can weakly increase (about 2-

fold) expression of the *Met* gene in BPH (**Figure 3G**). Our data furthermore indicates that *Met* and *FAMeT* expression levels were not drastically influenced by EB compared with *kr-h1* and *vg* (**Figure 3H and I**).

It should be mentioned that JH action does not directly result in the increase of *Met*. However, we cannot rule out the possibility that other factors (indirect effects), induced by EB treatment, increases the mRNA expression level of *Met*. One recent paper reported that downregulation of transcription factor *CncC* increases *met* expression in beetles (see **Figure 6A** in the reference) (Jiang et al., 2023). Furthermore, several studies have reported that insecticide treatment activates the *CncC* gene signaling pathway, which regulates detoxification gene expression (Amezian et al., 2023; Fu et al., 2024; Hu et al., 2021). Hence, it is possible that EB might influence the *CncC* gene pathway, which in turn induces *met* expression.

Notably, after exposing fourth-instar BPH nymphs to EB, no EB residues were detected in the subsequent adult stage. This finding indicates that the EB-induced increase in adult fecundity is initiated during the nymphal stage and manifests in adulthood - a mechanism distinct from the direct fecundity enhancement of fecundity observed when EB is applied to adults. We propose that sublethal EB exposure during critical nymphal stages may reprogram metabolic or endocrine pathways, potentially via insulin/JH crosstalk. For instance, increased nutrient storage (e.g., proteins, sugars; **Figure 2-figure supplement 2**) could enhance insulin signaling, which in turn promotes JH biosynthesis in adults (Ling and Raikhel, 2021; Mirth et al., 2014; Sheng et al., 2011). Future studies should test whether EB alters insulin-like peptide expression or signaling during development.

The EB-induced increase in fecundity in BPH is dependent on the allatostatin A signaling pathway

In addition to regulatory proteins that promote JH production, insects utilize peptides that regulate JH biosynthesis. These include the inhibitory allatostatins: FGLamides (FGLa; AstA), the W(X)6Wamides (AstB), and the PISCFs (AstC) (Nässel, 2002; Stay et al., 1996; Stay and Tobe, 2007; Tanaka et al., 2014; Wegener and Chen, 2022; Zhang et al., 2021). Interestingly, we found that EB exposure results in a marked downregulation of the expression of the genes encoding the allatostatin *AstA* and its receptor *AstAR*. We provide evidence for the functional impact of this on JH synthesis and BPH fecundity by: i) demonstrating that RNAi knockdown of *AstAR* expression results in increased JH titer and enhanced egg production, and, ii) showing that injection of female BPH with synthetic AstA peptide reduces the JH titer and decreases egg production. Thus, our data provide unequivocal evidence that AstA is a key inhibitor of JH production in BPH. This finding is consistent with previous work that has shown that FGLa/ASTs (AstA) inhibit JH biosynthesis in cockroaches, and termites (Woodhead et al., 1989; Yagi et al., 2005). To our knowledge, our study is the first report of insecticides down-regulating the expression of the neuropeptide receptor, *AstAR*, and linking this to increases in JH titer and enhanced reproduction in insects.

Interestingly knockdown of the *AstAR* resulted in significant increases in the expression of genes involved in JH biosynthesis/signaling including *JHAMT*, *Met* and *Kr-h1*. Related to this finding, previous work has shown that knockdown of the AstA receptor gene, *Dar-2*, in *D. melanogaster* results in changes in the expression of genes encoding *Drosophila* insulin-like peptides (DILPs) and adipokinetic hormone (AKH) (Hentze et al., 2015). Together with our findings, this demonstrates that AstA receptors may modulate the expression of numerous downstream genes involved in metabolism, energy storage and reproduction. In the case of pesticide resurgence our findings suggest that EB may act directly and/or indirectly on several genes that affect JH signaling and fecundity.

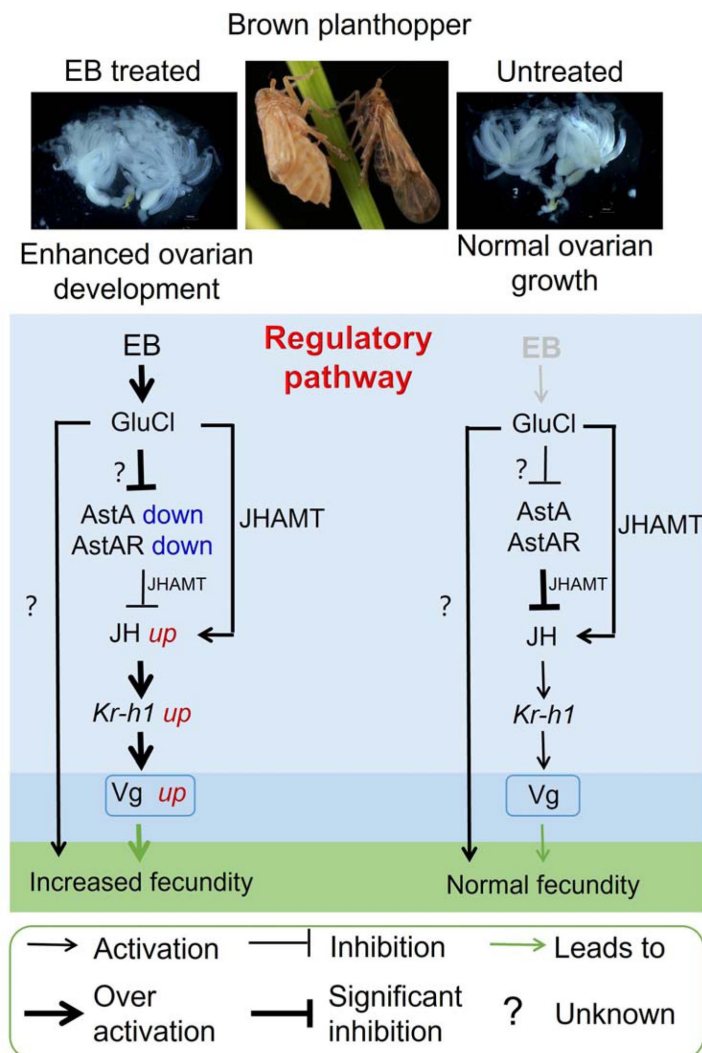


Figure 6.

Schematic of the proposed regulatory pathway of EB-enhanced fecundity in BPH.

Emamectin benzoate (EB) exposure results in the upregulation of genes that promote JH signaling pathway (*JHAMT* and *Kr-h1*) and the downregulation of genes that inhibit it (allatostatin, *AstA* and allatostatin A receptor, *AstAR*). This transcriptome reprogramming is dependent on the allosteric action of EB on its molecular target the glutamate-gated chloride channel (*GluCl*) receptor. Note that the mechanism of action on the *GluCl* is unknown in BPH, but likely the channel conformation changes and renders the receptor dysfunctional. Importantly, we do not suggest that the *GluCl* upregulation is due to direct action of EB on the channel. The resulting increased JH titer promotes vitellogenin (*vg*) biosynthesis and increased fecundity in EB exposed insects. We observe significant cross-talk in the expression of genes that inhibit JH production and those that promote it, with *AstAR* inhibiting the expression of *JHAMT*, *Met* and *Kr-h1* and *GluCl* activating the expression of *JHAMT* which is responsible for JH synthesis, and the JH signalling downstream genes *Met* and *Kr-h1*.

The GluCl plays a role in EB-induced fecundity in BPH

EB and abamectin are allosteric modulators of GluCl_s (Sparks et al., 2020 [↗](#); Sparks et al., 2021 [↗](#); Wu et al., 2017 [↗](#)). Our data revealed that EB exposure increases the expression of the GluCl in the adult stage of BPH, and knockdown of GluCl expression resulted in a reduction in both JH levels and egg production and egg laying. This EB action on *GluCl_α* expression is likely indirect, and we do not consider EB as transcriptional regulator of *GluCl_α*. Thus, the mechanism behind EB-mediated induction of *GluCl_α* remains to be determined. It is possible that prolonged EB exposure triggers feedback mechanisms (e.g. cellular stress responses) to counteract EB-induced *GluCl_α* dysfunction, leading to transcriptional upregulation of the channel. Hence, considering that EB exposure in our experiments lasts several days, these findings might represent indirect (or secondary) effects caused by other factors downstream of GluCl signaling that affect channel expression.

There are several studies showing that insects treated with insecticides display increases in the expression of target genes. For example, the relative expression level of the ryanodine receptor gene of the rice stem borer, *Chilo suppressalis* was increased 10-fold after treatment with chlorantraniliprole, an insecticide which targets the ryanodine receptor (Peng et al., 2017 [↗](#)). In *Drosophila*, starvation (and low insulin) elevates the transcription level of the receptors of the neuropeptides short neuropeptide F and tachykinin (Ko et al., 2015 [↗](#); Root et al., 2011 [↗](#)). In BPH, reduction in mRNA and protein expression of a nicotinic acetylcholine receptor $\alpha 8$ subunit is associated with resistance to imidacloprid (Zhang et al., 2015 [↗](#)). Knockdown of the $\alpha 8$ gene by RNA interference decreased the sensitivity of *N. lugens* to imidacloprid (Zhang et al., 2015 [↗](#)). Hence, the expression of receptor genes may be regulated by diverse factors, including insecticide exposure.

Interestingly, the GluCl has been reported to inhibit the biosynthesis of JH in the cockroach, *Diploptera punctata* (Liu et al., 2005 [↗](#)). Recent work has also shown that modulation of glutamatergic signals may contribute to the photoperiodic control of reproduction in bean bug, *Riptortus pedestris* (Hasebe and Shiga, 2022 [↗](#)). Furthermore, work on *D. punctata* has revealed that application of the GluCl channel agonist ivermectin, which like EB belongs to the avermectin family, caused a decline in JH synthesis in the corpus allatum (Liu et al., 2005 [↗](#)). While the inhibitory effect of ivermectin observed in this previous study differs from the activating effect of EB we observed in our study, it implicates a role for the GluCl channel in the regulation of JH regulation.

Conclusion

Our study has revealed that exposure of BPH to EB affects a diverse suite of genes that act in combination to enhance the JH titer and thus increase female fecundity. A schematic of how these factors promote ovarian maturation in the adult stage of *N. lugens* through the AstA and JH signaling pathway is provided in **Figure 6** [↗](#). Our findings provide the foundation for further work to understand how these genes interact and the mechanisms by which their expression is activated or repressed by EB. Furthermore, our findings provide fundamental insights into the molecular response in insects to xenobiotic stress and illustrate that pesticides can have unexpected and positive impacts on pest populations. In this regard our findings also have applied implications for the control of a highly damaging crop pest. Previous studies have reported that avermectins such as abamectin are toxic to the wolf spider *Pardosa pseudoannulata*, which is the main predator of BPH in rice crops (Chen et al., 2017 [↗](#); Sogawa, 2015 [↗](#)). Thus, these insecticides both stimulate reproduction in BPH and kill their natural enemies providing a ‘perfect storm’ for inducing damaging BPH outbreaks. Based on these findings, to avoid BPH resurgence, we suggest that the agents, EB and abamectin, should not be (or rarely be) applied to rice plants at growth

stages when BPH are present. On a more optimistic note, our findings have identified numerous genes that play key roles in BPH reproduction and thus represent promising targets for the development of novel controls against this important pest.

Materials and methods

Insects

BPH was initially collected from Wanan, JiangXi Province in 2020, reared on ‘Taichung Native 1’ (TN1) rice seedlings in the laboratory without exposure to any insecticides. The strain was maintained in a climatic chamber at $27 \pm 1^\circ\text{C}$, with relative humidity of $70 \pm 10\%$ and a light: dark = 16 h: 8 h photoperiod.

Chemicals

Emamectin benzoate (95.2%) was obtained from Hebei Weiyuan Co., Ltd. (Hebei, China). Abamectin (96.8%) was obtained from Hebei Weiyuan Co., Ltd. (Hebei, China). Pyriproxyfen (98%) was obtained from Shanghai Shengnong Co., Ltd. (Shanghai, China). Methoprene (S)-(+)(mx7457-100mg) was purchased from Shanghai MaoKang Biotechnology Co., Ltd., (Shanghai, China). Juvenile hormone standard sample (J912305-10mg) was purchased from Shanghai Macklin Biotechnology Co., Ltd., (Shanghai, China).

Bioassay

Different life stages of insects were used to perform bioassays to investigate the effects of insecticide on nymphs and adults. To test whether treatment of the nymph stage of insects would promote reproduction in female, we used 4th instar nymphs of BPH or 3rd instar nymph of *Laodelphax striatellus* and *Sogatella furcifera* to perform bioassays. To test whether treatment of the adult stage of insects would promote reproduction in female, we used newly emerged male and female BPH.

Systemic bioassays

The rice-seeding dipping bioassay method was used to evaluate the susceptibility of BPH, *L. striatellus* and *S. furcifera* to EB. Technical-grade insecticides were dissolved in acetone as stock solution then diluted in a series of six concentrations with water containing 0.1% Triton. Selected rice seedlings at the 6–8 cm growth stage were dipped in insecticide solutions for 30 s and then air-dried at room temperature. The roots of the rice seedlings were wrapped with cotton strips and seedlings placed in a plastic cup 5 cm in diameter. Fifteen insects were introduced into each plastic cup for each replicate. The top of the cup was then sealed with gauze to prevent escape. All experiments comprised at least three biological replicates. Control rice seedlings were treated with 0.1% Triton X-100 water solution only. All treatments were maintained under standard conditions of $27 \pm 1^\circ\text{C}$ and 70–80% relative humidity with a 16 h light/8 h dark photoperiod. Mortality was assessed after 4 d for *N. lugens* or 2 d for *L. striatellus* and *S. furcifera* after treatment with insecticides. The insects were considered dead if they were unable to move in response to gentle prodding with a fine brush.

For *Drosophila* larval bioassays, we adopted a method described previously in our lab with minor modifications (Huang et al., 2020 [DOI](https://doi.org/10.7554/eLife.91774.3)). Briefly, twenty third instar larvae were placed in fly vials containing fly food (based on corn powder, brown sugar, yeast and agar) supplemented with EB of different concentrations. Four concentrations (LC₁₀, LC₃₀ and LC₅₀) were tested together with a negative (no insecticide) control. For *Drosophila* adult bioassays, we selected virgin females three days after eclosion. Several concentrations were overlaid onto fly food in standard *Drosophila* vials and allowed to dry overnight at room temperature. A total of 15 adult flies (three days after

eclosion) were then added to each vial and mortality assessed after 2 d. Four replicates were carried out for each concentration. Control mortality was assessed using vials containing food with solvent minus insecticide.

Contact bioassays

For topical bioassays working insecticide solutions were prepared in acetone. 4th instar nymphs or newly emerged males/females were anesthetized with carbon dioxide for 5 s, and then 0.04 μ l/insect test solution applied topically to the dorsal plates with a capillary micro-applicator (Yao et al., 2017 [DOI](#)). Insects were then placed in an artificial climate incubator with a temperature of 27 $\pm$ 1°C, a photoperiod of 16:8 h (L:D), and a humidity of 70% $\pm$ 10%. Mortality was determined 2 d after treatment. Data with over 20% mortality in the control treatment were discarded, and the assay was repeated at least three times.

Fecundity assays

Fourth instar nymphs of BPH and 3rd instar nymph of *L. striatellus* and *S. furcifera* were treated with the LC₁₅ and LC₅₀ of EB, and then transferred to fresh rice seedlings. After eclosion, the adults were used in the following experiment. Newly emerged treated adults and untreated adults were paired to produce four groups: untreated males and untreated females (σ ck $\times$ φ ck; ck indicates untreated); untreated males and treated females (σ ck $\times$ φ t; t indicates insecticide treated); treated males and untreated females (σ t $\times$ φ ck); treated males and treated females (σ t $\times$ φ t). Each group comprised at least 10 mating pairs. All pairs were transferred to glass cylinders (diameter 5 cm and height 20 cm) containing rice plants (25 days old almost 20 cm high) as a food source for eleven days. The number of eggs and nymphs in plants were counted under a microscope.

For *Drosophila* egg-laying assays, we adopted our previous method (Wu et al., 2019 [DOI](#)). Briefly, insecticide-treated virgin females were paired with untreated males for three days and then the mated females transferred into the *Drosophila* ovipositing apparatus. The number of eggs were counted after 16 hours.

Fitness analysis

The fitness of EB-treated BPH were analyzed using methods reported previously (Zeng et al., 2023 [DOI](#)). We selected two groups, (σ ck $\times$ φ ck) and (σ ck $\times$ φ t), to study the effects of the LC₅₀ concentration of EB on BPH fitness. In the case of systemic exposure, 4th instar nymphs of BPH were treated with the LC₅₀ of EB for 4 days and then transferred to tubes containing untreated rice plants for individual rearing. The rice plants were replaced every three days with untreated plants. The emergence ratio, female ratio, preoviposition period, female longevity, brachypterism female ratio and female weight were calculated.

Examination of ovarian development

Adult females from σ ck $\times$ φ ck control and σ ck $\times$ φ t group on 1, 3, 5, 7 DAE were dissected to observe ovarian development. The mature eggs in the ovary were photographed and recorded. Each group comprised at least fifteen replicates.

To examination whether EB treatment impairs egg maturation, we dissected untreated or EB-treated ovaries and fixed them in 4% paraformaldehyde (PFA) in phosphate buffered saline (PBS) for 30 min at room temperature. After four washes of 10 min (4 $\times$ 10 min) in PAT3 (PBS with 0.5% Triton X-100 and 0.5% bovine serum albumin), the ovaries were then incubated with DAPI (4',6-diamidino-2-phenylindole, 100 nM) and Actin-stain 670 Fluorescent Phalloidin (200 nM). Imaging was performed using a Zeiss LSM980 confocal laser microscope.

Measurements of glycogen, triglyceride, total protein content, cholesterol and four sugars

The content of glycogen, triglyceride, cholesterol and total protein was determined by spectrophotometry at 620 nm, 510 nm, 510 nm and 562 nm respectively using the glycogen assay kit (A043-1-1), triglyceride reagent kit (A110-1-1), cholesterol assay kit (A111-1-1) and total protein assay kit (A045-2-2) obtained from Nanjing Jiancheng Bioengineering Institute following the manufacturer's instructions. The determined results were normalized to the protein content in the sample, which was determined using BCA Protein Assay Reagent Kit (Thermo Scientific, Waltham, USA). Each sample contained tissue extracts from five adult female BPH, with three biological replicates per sample.

To assess the content of four sugars (sucrose, glucose, fructose and trehalose) in the extract of BPH tissue, the same extraction method was used as above. Sugar content was quantified using the colorimetric method by scanning at 290 nm, 505 nm, 285 nm and 620 nm respectively using the sucrose reagent kit (A099-1-1), glucose reagent kit (F006-1-1), fructose reagent kit (A085-1-1) and trehalose reagent kit (A150-1-1) obtained from Nanjing Jiancheng Bioengineering Institute based on the manufacturer's instructions. Each sample contained tissue extracts from five adult female *N. lugens*, with three biological replicates per sample.

Determination of Juvenile hormone III and 20-hydroxyecdysone titers of BPH by ELISA

The titer of Juvenile hormone III in BPH was measured using the Juvenile hormone ELISA Kit (Lot Number: HLE92086, Shanghai HaLing Biotechnology Co., Ltd., Shanghai, China) which employs the dual-antibody sandwich ELISA method. The titer of 20-hydroxyecdysone in BPH was measured using the 20-hydroxyecdysone ELISA Kit (Lot Number: ZK14705, Shanghai ZhenKe Biotechnology Co., Ltd., Shanghai, China). At least three biological replicates were employed for each treatment.

Determination of Juvenile hormone III titer in BPH using HPLC-MS/MS

The whole bodies of 5 individual BPH were mixed with 1 ml of n-hexane, followed by centrifugation at 10,000×g for 10 min, the upper hexane layer was then dried with nitrogen, dissolved in methanol and sonicated for 10 min, after centrifugation at 10000×g for 10 min, the supernatant was collected through the organic filter membrane of 0.22 µm into 2 ml vials for JH LJ determination. JH LJ standard sample (J912305-10 mg) purchased from (Shanghai McLean Biochemical Technology Co. Ltd), dissolved in methyl alcohol as stock solution 10,000 mg/L was diluted in a series of six concentration gradients to serve as a reference. Liquid chromatography-tandem mass spectrometry (LC-MS/MS) was then carried out using UPLC Xevo TQ-S Micro (Waters technology), quantitative method according to the external standard, the chromatographic column was EC-C18 (4.6 mm×150 mm, 2.7 µm), column temperature was 30°C injection volume was 20 µl, elution flow rate was 0.3 ml/min, and the mobile phase was acetonitrile:formic acid water (90:10), detection wavelength was 218 nm, the peak height was used for quantification.

Determination of EB content using HPLC-MS/MS

4L BPH were systemically treated with EB as outlines above, samples were collected at 3 days post-treatment and transferred to healthy rice seedlings for 3 more days, after which another sample was collected. Each experiment included 3 biological replicates, with approximately 20 brown planthoppers weighed per replicate. The weight of each sample was recorded, and 400 µL of methanol was added to each sample. The samples were then ground for 180 seconds at 65 Hz, followed by ultrasonic treatment at 4°C for 30 minutes. Next, 400 µL of n-hexane was added, and the mixture was vortexed to mix thoroughly. The samples were centrifuged at 12,000 rpm for 15

minutes at 4°C, and the supernatant was collected into a centrifuge tube and evaporated to dryness in a concentration evaporator. Then, 200 µL of methanol was added for reconstitution, followed by vortex mixing. The samples were centrifuged again at 12,000 rpm for 15 minutes at 4°C, and the supernatant was analyzed using the LC-MS/MS system. The EB standard was purchased from Shanghai Minxin Biotechnolog. Six concentration gradients were used for elution. The column temperature was maintained at 40°C, and the flow rate was set to 0.300 mL/min. Liquid chromatography-tandem mass spectrometry (LC-MS/MS) was performed using a Waters Acquity UPLC system and an AB SCIEX 5500 QQQ-MS mass spectrometer. The chromatographic column used was Acquity UPLC BEH C18 (1.7 µm, 2.1 mm × 100 mm). A standard curve was generated for content calculation.

Cloning, sequence and phylogenetic analysis

The NCBI database and BLAST program were used to carry out sequence alignment and analysis. Open Reading Frames (ORFs) were predicted with EditSeq. Primers were designed using the primer design tool in NCBI. Total RNA Extraction was extracted from 30 adults BPH using TRIzol reagent (Invitrogen, Carlsbad, CA, USA) according to the manufacturer's instructions. cDNA was synthesized using the Biotech M-MLV reverse transcription kit. Full-length gene sequences were amplified by PCR using cDNA as template and Rapid Taq Master Mix (Vazyme Biotech, Cat# P222-02). The PCR product was purified on a 1% agarose gel, cloned into pClone007 Simple Vector Kit (Tsingke Biotech, Cat# TSV-007S), and then sequenced using the 3730 XL DNA analyzer (Applied Biosystems, Carlsbad, CA, USA). Table S2 contains a list of the primers used in this study.

The exon and intron architectures of *AT*, *AstA*, *AstB*, *AstCC* and *AstACCC* were predicted based on the alignments of putative ORFs against their corresponding genomic sequences. Sequence similarity/annotations and orthologous gene searches were performed using BLAST programs available in NCBI. Multiple alignments of the complete amino acid sequences were performed with Clustal Omega (<http://www.ebi.ac.uk/Tools/msa/clustalo>). Phylogeny was conducted using the maximum likelihood technique to create phylogenetic trees and these were bootstrapped with 1000 replications were used using the MEGA 6 software (Tamura et al., 2013).

RNA interference

Double-stranded RNA (dsRNA) of *gfp* (green fluorescent protein), *JHAMT* (juvenile hormone acid O-methyltransferase), *Met* (methoprene-tolerant), *Kr-h1* (krüppel homolog 1), *AstAR* (allatostatin-A receptor) and *GluCl* (glutamate-gated chloride channel) was prepared using Ambion's MEGAscript T7 kit following the manufacturer's instructions. The primer sequences for double-stranded RNA synthesis are listed in Table S2. Newly emerged females were injected with 40 nl (5,000 ng/µl) of double-stranded RNA of *gfp* (*dsghp*) or double-stranded RNA of the target genes in the conjunctive part between prothorax and mesothorax of insects. In the RNAi experiments, BPH were then treated with the LC₅₀ of EB 24h after dsRNA injection and the whole body sampled for qRT-PCR analysis.

Quantitative RT-PCR

Fourth instar nymphs of BPH were treated with EB after which total RNA was extracted from 5th instar nymphs and 1-7 day post-eclosion females of *N. lugens* using the methods detailed above. The HiScript® II RT SuperMix for qPCR (+gDNA wiper) kit from Vazyme, Nanjing, China, was used to create first-strand cDNA. Primer3 (<http://bioinfo.ut.ee/primer3/>) was used to design real-time quantitative PCR (qPCR) primers listed in Table S2. mRNA levels of candidate genes were detected by qPCR using the UltraSYBR Mixture (with ROX) Kit (CW BIO, Beijing, China). Each reaction contained 2 µL of cDNA template (500 ng), 1 µL each forward and reverse qPCR primer (10 µM), 10 µL of UltraSYBR mixture buffer, and 6 µL of RNase-free water. Q-PCR was run on an ABI 7500 Real-Time PCR System (Applied Biosystems, Foster City, CA) under the following conditions: 5 min at 95°C, followed by 40 cycles of 95°C for 10 s and 60°C for 30 s. Three independent biological

replicates and four technical replicates were used in each qPCR experiment. The housekeeping gene 18S ribosomal RNA was used to normalize the expression of candidate genes. The $2^{-\Delta\Delta Ct}$ method (Ct represents the cycle threshold) was used to measure relative expression levels (Livak and Schmittgen, 2001 [↗](#)). Three biological replicates were used for statistical comparison between samples. Table S2 contains a list of the primers used in this study.

Gene expression quantification following RNAi

In late fourth-instar nymphs, we injected 40 nl of synthesized dsRNA targeting *GFP*, *JHAMT*, or *Kr-h1* (5,000 ng/μl). After a 24-hour recovery period, we applied emamectin benzoate (EB) using the systemic route described previously, while control rice seedlings were treated with a 0.1% Triton X-100 aqueous solution. Seventy-two hours post-treatment, we collected female brown planthoppers from each treatment group (at least four groups, $n \geq 5$ per group) for total RNA extraction and first-strand cDNA synthesis. We then performed qPCR to measure mRNA levels of the candidate genes *JHAMT* and *Kr-h1* using the UltraSYBR Mixture (with ROX) kit (Beijing Zhongwei Biotech Co., China). The primers for dsRNA synthesis and the qPCR primers were identical to those used in the previous experiments.

Statistics

PoloPlus v2.0 (LeOra Software 2008) was used to calculate the lethal concentration (LC₅₀) and 95% fiducial limits (95% F.L.) from bioassay data. GraphPad Prism 8.0 software (GraphPad Software Inc., San Diego, USA) was used to generate graphs and perform statistical analysis of data. Data presented in this study were first verified for normal distribution using the D'Agostino–Pearson normality test. One-way analysis of variance (ANOVA) with Duncan's multiple range test was used to test differences among multiple groups of normally distributed groups of data. Student's t test was used to test the differences between two groups. If data was not normally distributed, Mann–Whitney tests were used for pairwise comparisons, and Kruskal–Wallis test used for comparisons among multiple groups, followed by Dunn's multiple comparisons. All data are presented in figures as mean ± s.e.m. The sample sizes and statistical tests used for each experiment are stated in the figures or figure legends.

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Additional information

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National Natural Science Foundation of China (32472542)

Figure 1-figure supplement 1.

(A) Fecundity of BPH when newly emerged adults were treated with sub-lethal (LD_{15}) and median lethal (LD_{50}) concentrations of emamectin benzoate via topical application. (B) Fecundity of BPH when 4th instar nymphs were treated with sub-lethal (LC_{15}) and median lethal (LC_{50}) concentrations of abamectin via systemic exposure. (C) Residual levels of emamectin benzoate (EB) in brown planthoppers (BPH) following treatment. Nymphs (4th instar) were exposed to EB, and residues were quantified by HPLC-MS/MS method in nymphs (immediately post-treatment) and adults (after eclosion). No detectable EB residues were observed in adults ($P=0.0037$, two-tailed t-test), indicating complete metabolic clearance or degradation during development. All data are presented as the mean $\pm$ s.e.m. Different lower-case letters above the bars indicate significant differences (One-way ANOVA with Tukey's Multiple Range Test, $p < 0.05$).

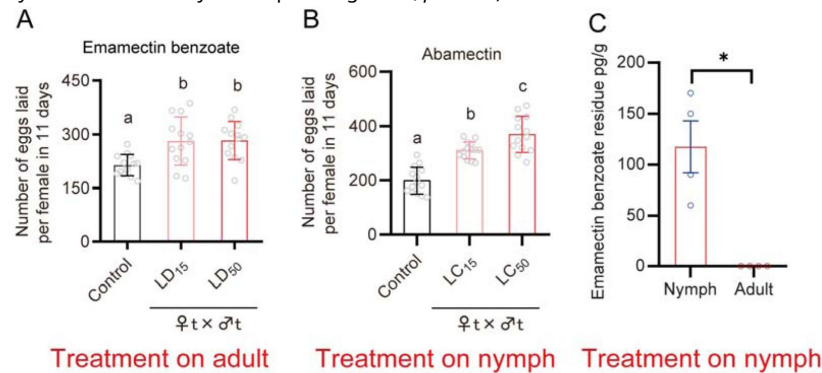
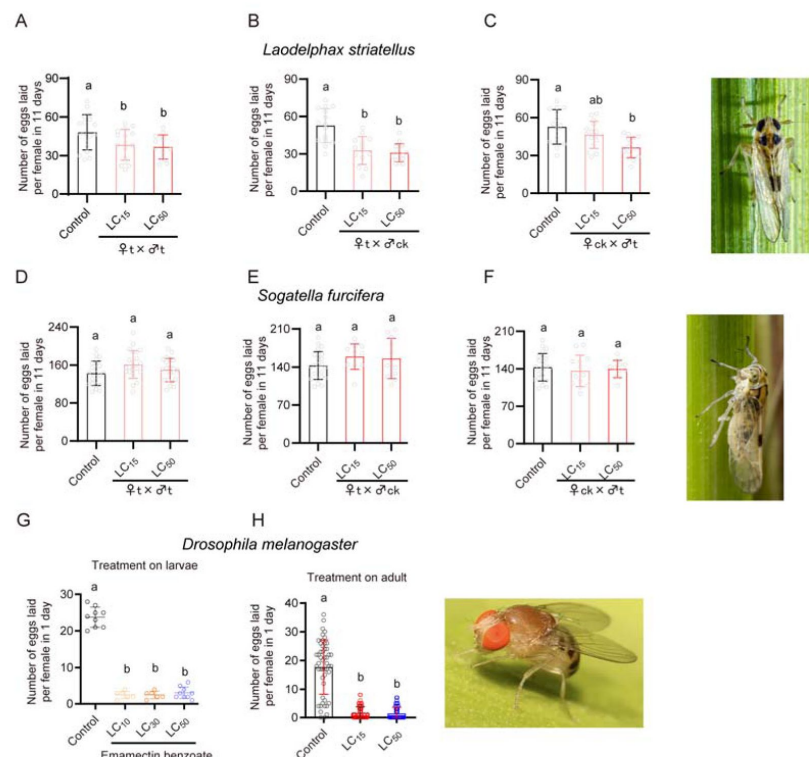


Figure 1-figure supplement 2.

Fecundity of small brown planthopper, *Laodelphax striatellus*, (A-C) white backed planthopper, *Sogatella furcifera* (D-F) and vinegar fly, *Drosophila melanogaster* (G and H) when larvae and newly emerged adults were treated with sub-lethal concentrations of emamectin benzoate.

All data are presented as the mean $\pm$ s.e.m. Different lower-case letters above the bars indicate significant differences (One-way ANOVA with Tukey's Multiple Range Test, $p < 0.05$).



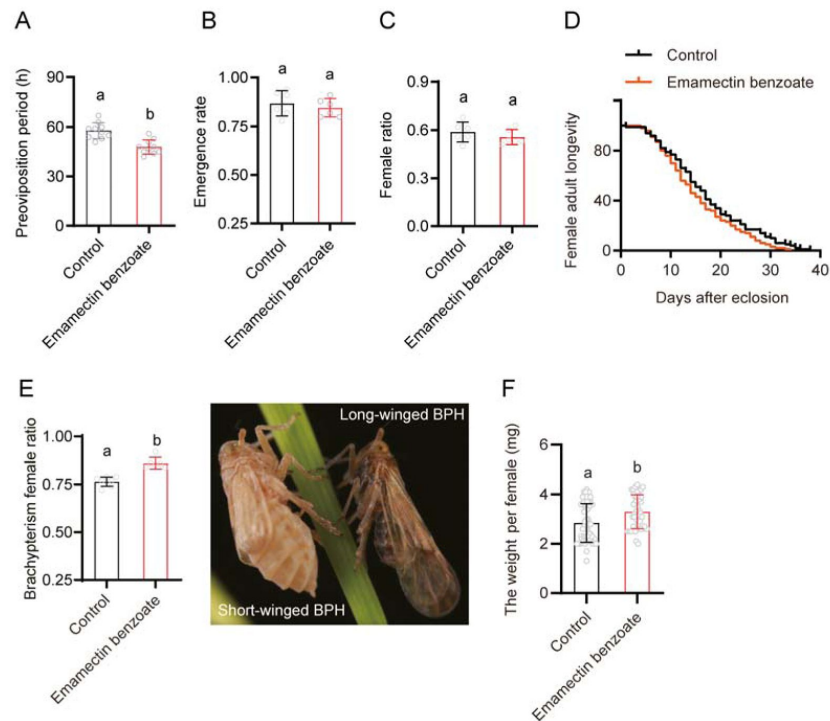


Figure 1-figure supplement 3.

The impact of emamectin benzoate on the reproductive fitness of BPH.

Fourth instar nymphs were treated with the LC_{50} concentration of emamectin benzoate in systemic bioassays. (A) Preoviposition period: Preoviposition refers to the period in an insect's life cycle between the time it becomes an adult and the time it starts laying eggs. (B) Emergence rate: the rate of emergence of adults; (C) Female ratio: the ratio of female (to male) insects in the total of emerged adults; (D) Female adult longevity. (E) Brachypterism female ratio: the ratio of short-winged to long-winged adults; (F) The weight per female. All data are presented as the mean $\pm$ s.e.m. Different lower-case letters above the bars indicate significant differences (Student's t test, $p < 0.05$).

Figure 2-figure supplement 1.

Number of mature eggs in the ovaries of EB treated BPH female adults compared to controls.

The data are presented as the mean $\pm$ s.e.m. Asterisks indicate values significantly different from the control using student t test ($*p < 0.05$).

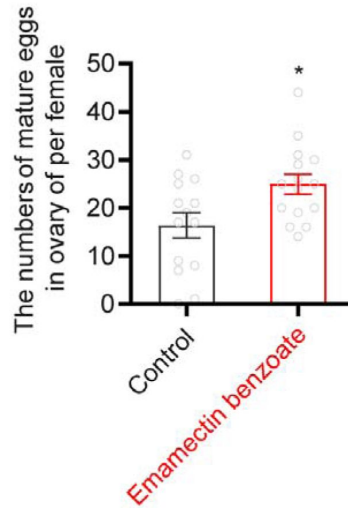


Figure 2-figure supplement 2.

The oogenesis of BPH.

See the main text for detailed information.

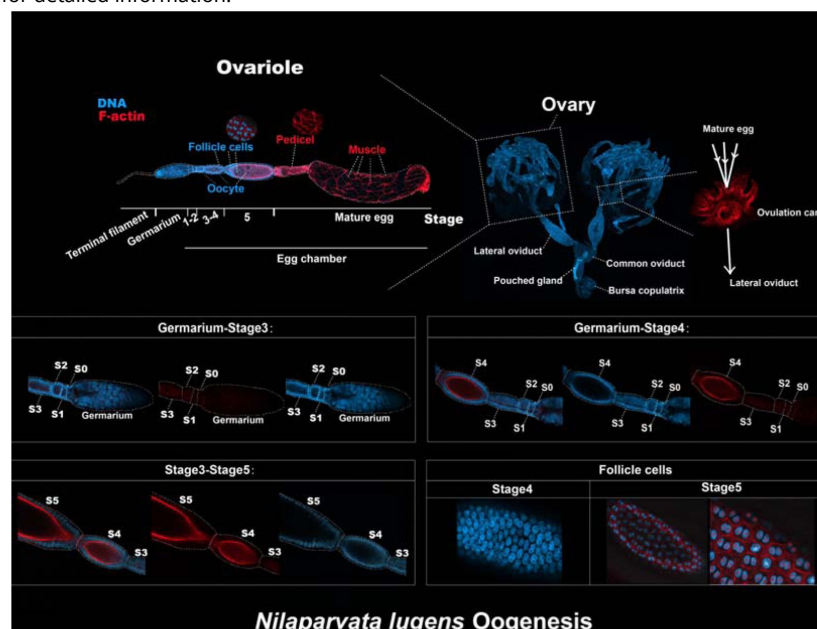


Figure 2-figure supplement 2.

Amounts of Glycogen (A), TAG (B), total protein content (C), cholesterol (D) and four circulating sugars including sucrose, glucose, fructose and trehalose (E-H) after BPH exposure to EB.

All data are presented as the mean $\pm$ s.e.m. The differences between the EB-treated and solvent-treated BPH were analyzed using unpaired student *t*-test (*, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$; ****, $p < 0.0001$).

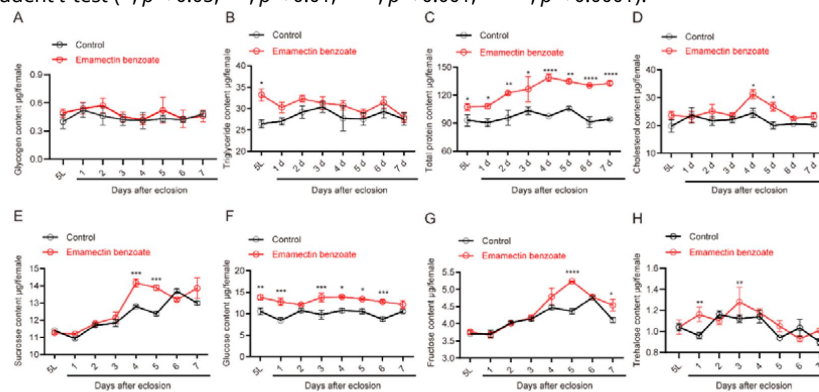


Figure 3-figure supplement 1.

20-hydroxyecdysone titer at different developmental stages of whole body BPH when 4th instar nymphs were treated with median lethal concentrations of EB.

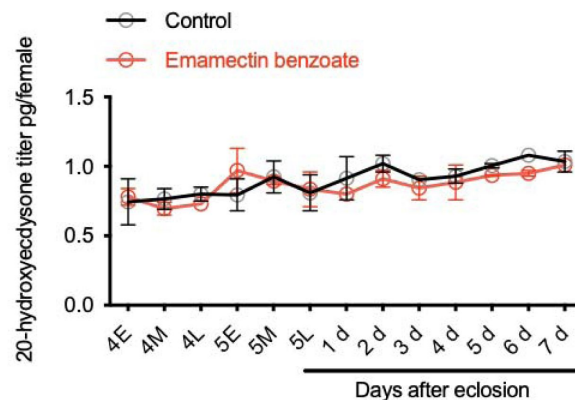


Figure 3-figure supplement 2.

(A) Expression of *Kr-h1* following RNAi knockdown. (B) Expression of *Vg* when *Kr-h1* was silenced in BPH. All data are presented as means $\pm$ s.e.m. * $p < 0.05$; Mann-Whitney test.

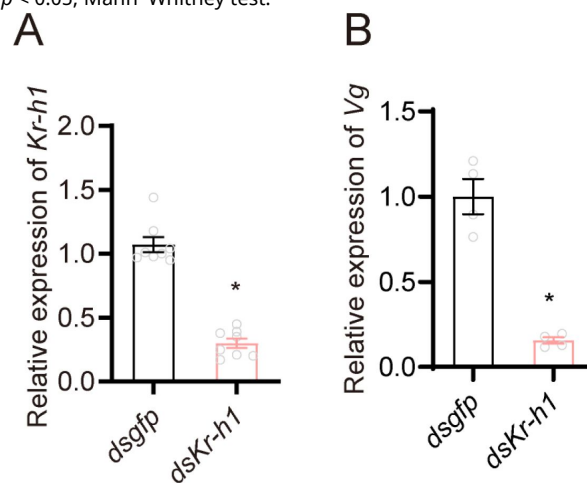
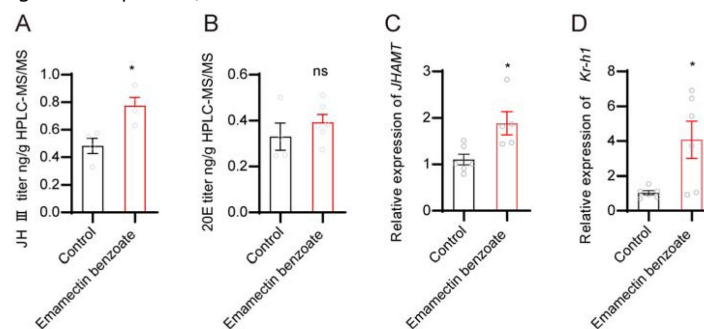


Figure 3-figure supplement 3.

(A) The titer of JH III (as measured by HPLC-MS/MS) in BPH females at adult stages after treatment with median lethal concentrations of EB. (B) The titer of 20-hydroxyecdysone in BPH when female adults were treated with median lethal concentrations of EB. (C and D) Expression of *JHAMT* and *Kr-h1* in EB-treated BPH female adults. All data are presented as means $\pm$ s.e.m. Student's *t* test was used to compare controls and treatments. Asterisks indicate values significantly different from the control (ns, no significant; * $p < 0.05$).



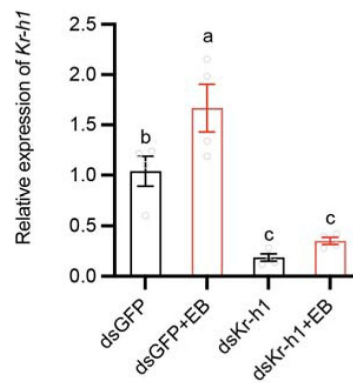


Figure 3-figure supplement 4.

Evaluation of the potential of EB to reverse dsRNA-mediated silencing by quantifying *Kr-h1* gene expression.

Different lower-case letters above the bars indicate significant differences (One-way ANOVA with Tukey's multiple comparisons test, $p < 0.05$).

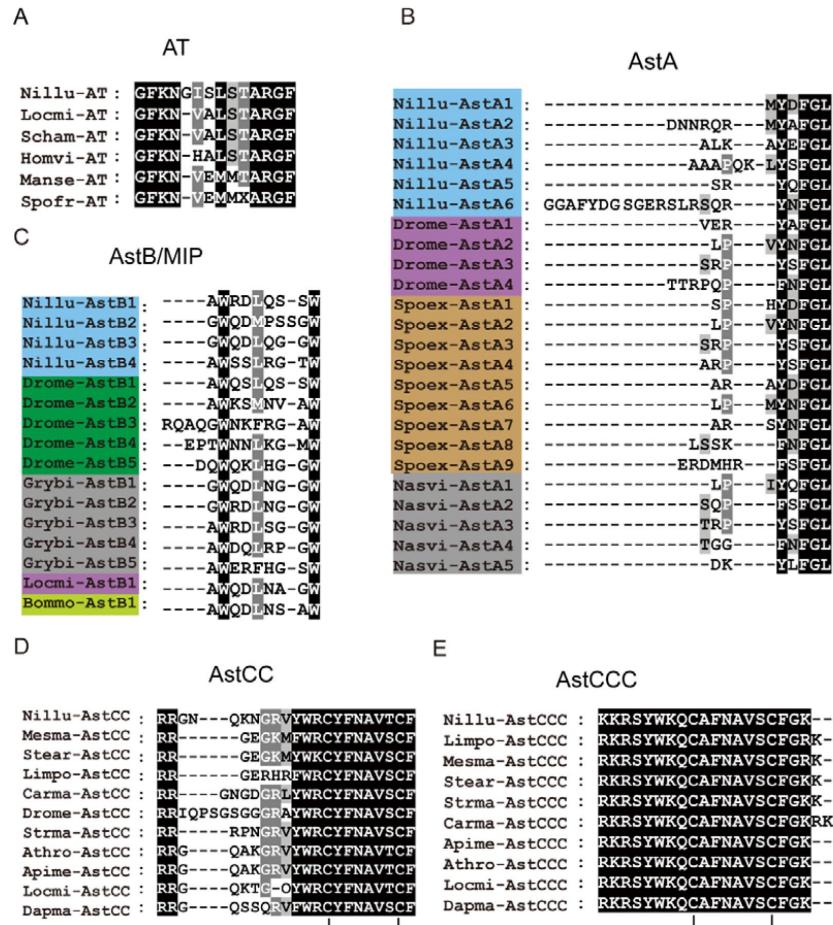


Figure 4-figure supplement 1.

Alignments of the amino acid sequences of: (A) AT, (B) AstA, (C) AstB/MIP, (D) AstCC and (E) AstCCC peptides from select species.

AT, AstA, AstB/MIP and AstCCC are predicted to have a C-terminal amide. The mature peptides belonging to the same species have been highlighted with the same color. Species names are as follows: Nillu (*Nilaparvata lugens*), Locmi (*Locusta migratoria*), Scham (*Schistocerca americana*), Homvi (*Homalodisca vitripennis*), Manse (*Manduca sexta*), Spofr (*Spodoptera frugiperda*), Drome (*Drosophila melanogaster*), Spoex (*Spodoptera exigua*), Nasvi (*Nasonia vitripennis*), Grybi (*Gryllus bimaculatus*), Bommi (*Bombyx mori*), Mesma (*Mesobuthus martensii*), Stear (*Stegodyphus araneomorph*), Limpo (*Limulus polyphemus*), Carma (*Carcinus maenas*), Strma (*Strigamia maritima*), Athro (*Athalia rosae*), Apime (*Apis mellifera*), Dapma (*Daphnia magna*). Black lines under the sequences indicate the locations of the disulfide bridges in the mature peptides. The accession numbers of the sequences are listed in Figure 4-figure supplement 1 source data.

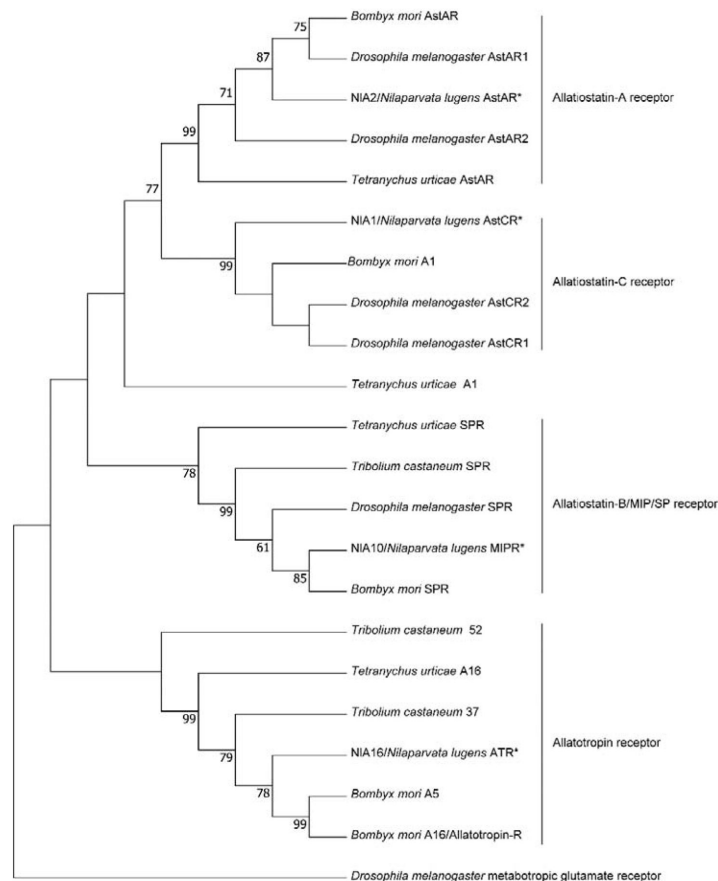


Figure 4-figure supplement 2.

Phylogenetic tree of the predicted BPH (*) allatotropin receptor (A16, ATR), allatostatins A receptor (A2, AstAR), AstB (MIP) receptor (A10, AstBR or MIPR) and allatostatins C receptor (A1, AstCR) with other insect species.

The tree was generated using the maximum likelihood method. *Drosophila melanogaster* metabotropic glutamate receptor was included as an outgroup. The accession numbers of the sequences used for this phylogenetic tree are listed in Figure 4-figure supplement 2 source data.

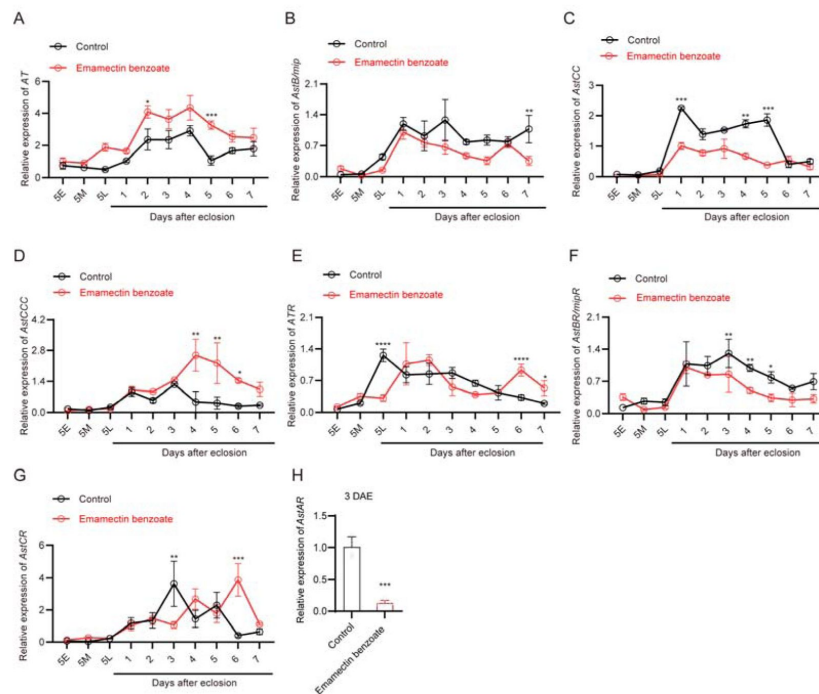


Figure 4-figure supplement 3.

EB induced changes in the expression of *AT*, *AstB*, *AstCC*, *AstCCC*, *ATR*, *AstBR* and *AstCR* in BPH.

All data are presented as means $\pm$ s.e.m. Student's t test was used to compare controls and treatments. ns, no significant difference; Asterisks indicate values significantly different from the control (* $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$).

Figure 5-figure supplement 1.

Phylogenetic analysis of glutamate-gated chloride channels in different species.

The numbers at the nodes of the branches represent the percentage bootstrap support (1000 replications) for each branch. The *Sogatella furcifera* GABA-gated chloride channel and *Nilaparvata lugens* nAChR α 6 were used as outgroup. Alignment was performed with amino acid sequences from TM1-7. The receptor names are listed in the tree. The accession numbers of the sequences used for this phylogenetic tree are listed in Figure 5-figure supplement 1 source data.

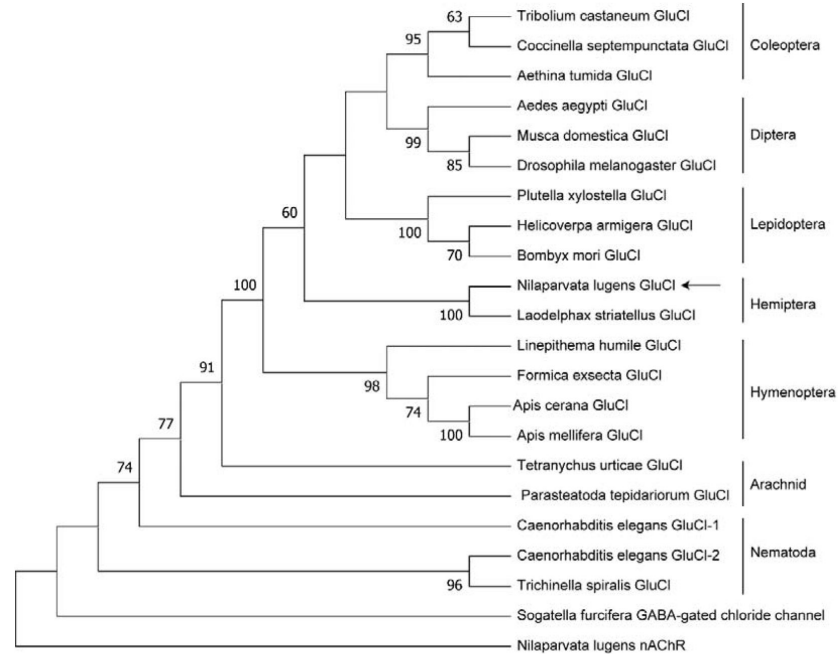


Figure 5-figure supplement 2.

The expression of *AT*, *AstA*, *AstB*, *AstCC*, *AstCCC*, *AstAR*, *AstBR* and *AstCR* in BPH injected with *dsGluCl* or *dsgfp*.

All data are presented as means $\pm$ s.e.m. Student's *t* test was used to compare the two samples. ns, no significant. Different lower-case letters above the bars indicate significant differences ($p < 0.05$).

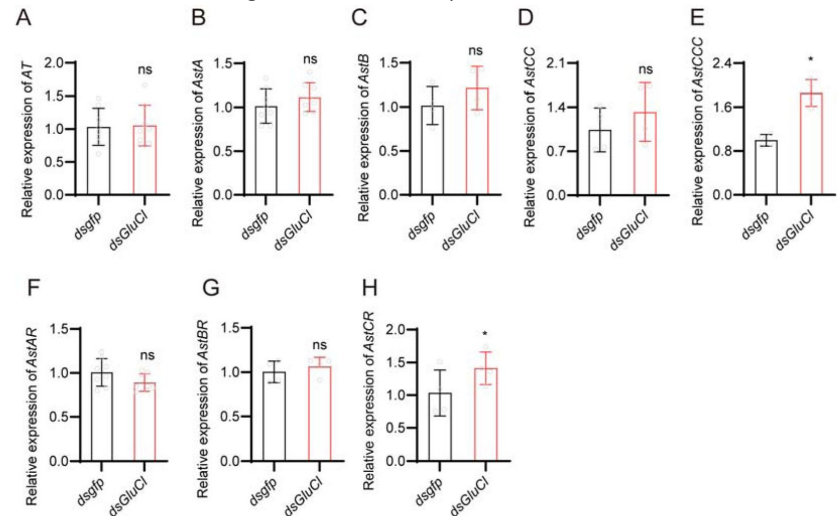


Figure 5-figure supplement 3.

The expression of *GluCl* after female adult BPH were treated with EB.

All data are presented as means $\pm$ s.e.m. Student's *t* test was used to compare the two samples, *, $p < 0.05$.

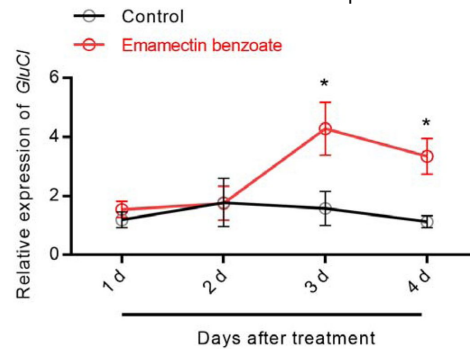


Table S1.

Determination of the toxicity of emamectin benzoate on BPH in systemic and topical application bioassays.

Treatment method	Developmental stages	Slope $\pm$ SE	LC ₁₅ (95%F.L.) (mg/L) or LD ₁₅ (95%F.L.) (pg/insect)	LC ₅₀ (95%F.L.) (mg/L) or LD ₅₀ (95%F.L.) (pg/insect)	χ^2 (df)	<i>P</i> value
Systemic route	4 th nymph	2.56 $\pm$ 0.487	0.87 (0.39-1.32)	2.21 (1.53-2.82)	2.68 (4)	0.61
	Adult Male	2.42 $\pm$ 0.35	0.80 (0.40-1.20)	2.13 (1.48-2.78)	2.12 (4)	0.72
	Adult Female	2.47 $\pm$ 0.41	2.33 (1.09-3.56)	6.12 (4.18-8.00)	3.56 (4)	0.50
Contact route	4 th nymph	2.52 $\pm$ 0.35	15.54 (9.46-21.42)	39.90 (30.66-49.56)	2.60 (4)	0.63
	Adult Male	2.77 $\pm$ 0.57	6.72 (2.52-10.92)	15.96 (9.66-21.84)	1.23 (3)	0.75
	Adult Female	3.31 $\pm$ 0.76	18.48 (6.72-28.14)	37.80 (23.10-50.40)	0.79 (3)	0.85

Primers	Primer sequences (5' - 3')
For cDNA cloning	
Allatotropin :	
NIAT-F	TACGCGGCCAAACACACTTA
NIAT-R	AGGGAAAGAGGGCGAAATTCA
Allatostatins :	
NIAsA-F	TCGGCCGTCACAAGTCAAG
NIAsA-R	CCGAACCCGTACTTCATGCT
NIAsB-F	ACCGGGCTCACAGGAATTTT
NIAsB-R	TGTAGGCGCAGATCTTGAGG
NIAsCC-F	AACACAGCTCTACGAGGCAC
NIAsCC-R	CCAAGCAGGTGACTGCCATA
NIAsCCC-F	TTTGTGTGTGCTTGCAAGTG
NIAsCCC-R	GGATAGAAACGGTAGATTGGTAGA
Allatostatins receptor :	
NIA16-F	CCTCATTGTGGAACACCGA
NIA16-R	CGCAGCTGTAAGGTGGAAGA
NIA2-F	GAACGTAATGGGAGTCGGCA
NIA2-R	GTTTTTGTGAGCGCCGACTT
NIA10-F	ATGCAAAACACGGCCAGCCT
NIA10-R	TTAATCGTCTCTGCTCAACTCCAAAGGAAGGT
NIA1-F	CGACCAGACCACTCTACTGC
NIA1-R	ACGTGGACCTCACTATACCAAAAA
For Quantitative RT-PCR	
Q-NI18S-F	CGCTACTACCGATTGAA
Q-NI18S-R	GGAAACCTTGTTACGACTT
Q-vitellogenin-F	GTGGCTCGTTCAAGGTTATGG
Q-vitellogenin-R	GCAATCTCTGGGTGCTGTTG
Q-Vitellogenin receptor-F	AGGCAGCCACACAGATAACCGC
Q-Vitellogenin receptor-R	AGCCGCTCGCTCCAGAACATT
Q-JHE-F	GAGCCTCACATCCACAGC
Q-JHE-R	AATGGGAGCCCTACGC
Q-NIMet-F	GGTGGTAAACGGATTGGAAG
Q-NIMet-R	CATCGTCAGCCAACTCGATA
Q-JHAMT-F	GAACCTGCAGGCCAAACACA
Q-JHAMT-R	ACCACTCGGTTGGGCTGAAT

Table S2

Sequences of oligonucleotide primers used in this study.

Q-NIKr-h1-F	TGATGAGGCACACGATGACT
Q-NIKr-h1-R	ATGGAAGGCCACATCAAGAG
Q-NIAT-F	CACGATACGTGGCTTCAAGA
Q-NIAT-R	ACGATCACTTTCGCCAATTC
Q-NIAstA-F	AGGACTTACTGGGCGAGGAT
Q-NIAstA-R	GGTGTCTCGTTTCCTGGTGT
Q-NIAstB-F	AGCGAGCTAGACGAGGACAA
Q-NIAstB-R	TCGTCTCTGCTCAACTCCAA
Q-NIAstCC-F	CTGCTCCCAGTGAAAAGGAG
Q-NIAstCC-R	GCTTCCAGTAACTGCGCTTC
Q-NIAstCCC-F	TGTGCTTGCAGGTGGTAGTC
Q-NIAstCCC-R	AGAAGCATGTGACTGCGTTG
Q-NIA2-F	TCCTGGTGTGAAGAGTGTG
Q-NIA2-R	CTTTTCGGGCCCCATTAATTT
Q-NLA1-F	ATATCGGCACCGAAGATGAG
Q-NLA1-R	GTCTGACCCGACAGGTTCTC
Q-NLA10-F	ACTGGGTGTCGACCAATCTC
Q-NLA10-R	TCGGTAGCGAGGAAGACAGT
Q-NLA16-F	TACCGTTCTGTGGGATGTCA
Q-NLA16-R	CCGGGATATCAAAGACGAGA
Q-NIGluCI-F	CACTGACTGAGGCCAACAGA
Q-NIGluCI-R	GCTGGCCATTCTTAGTGAGC

For double-stranded RNA synthesis

T7-EGFP-F	TAATACGACTCACTATAGGGCGTAAACGGCCACAAGTTCA
T7-EGFP-R	TAATACGACTCACTATAGGGGACTGGGTGCTCAGGTAGTG
T7-Kr-h1-F	TAATACGACTCACTATAGGGCGCCAGTGAAAGTGAGACCT
T7-Kr-h1-R	TAATACGACTCACTATAGGGGAGACCGCAAGTGTTCTGA
T7-Met-F	TAATACGACTCACTATAGGGCCACCAACCAGCAGATGAACCTGA
T7-Met-R	TAATACGACTCACTATAGGGCCACGCAAAGCCTCGTACTCTTGG
T7-JHAMT-F	TAATACGACTCACTATAGGGCTCCAGGCCATTGTCCCTCA
T7-JHAMT-R	TAATACGACTCACTATAGGGTTGGCCTGCAGGTTCTTTGG
T7-AstA-R-F	TAATACGACTCACTATAGGGTACTGCCGTTCTGGCCTTTT
T7-AstA-R-R	TAATACGACTCACTATAGGGGTGATCTGGAAGAGCGGCTT
T7-GluCI-F	TAATACGACTCACTATAGGGACACATCACCTGCTCACCTG
T7-GluCI-R	TAATACGACTCACTATAGGGGTGTGTTGCCTGCTGCTG

Table S2 (continued)

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Reviewer #1 (Public review):

Summary:

Gao et al. has demonstrated that the the pesticide emamectin benzoate (EB) treatment of brown plathopper (BPH) leads to increased egg laying in the insect, which is a common

agricultural pest. The authors hypothesize that EB upregulates JH titer resulting in increased fecundity.

Strengths:

The finding that a class of pesticide increases fecundity of brown planthopper is interesting.

Comments on revisions:

All my concerns have been addressed to reasonable level of satisfaction.

<https://doi.org/10.7554/eLife.91774.3.sa1>

Author response:

The following is the authors' response to the previous reviews.

Reviewer #1 (Recommendations for the authors):

(1) The onus of making the revisions understandable to the reviewers lies with the authors. In its current form, how the authors have approached the review is hard to follow, in my opinion. Although the authors have taken a lot of effort in answering the questions posed by reviewers, parallel changes in the manuscript are not clearly mentioned. In many cases, the authors have acknowledged the criticism in response to the reviewer, but have not changed their narrative, particularly in the results section.

We fully acknowledge your concern regarding the narrative linking EB-induced GluCl expression to JH biosynthesis and fecundity enhancement, particularly the need to address alternative interpretations of the data. Below, we outline the specific revisions made to address your feedback and ensure the manuscript's narrative aligns more precisely with the experimental evidence:

(1) Revised Wording in the Results Section

To avoid overinterpretation of causality, we have modified the language in key sections of the Results (e.g., Figure 5 and related text):

Original phrasing:

“These results suggest that EB activates GluCl which induces JH biosynthesis and release, which in turn stimulates reproduction in BPH (Figure 5J).”

Revised phrasing:

“We also examined whether silencing Glucl α impacts the AstA/AstAR signaling pathway in female adults. Knock-down of Glucl α in female adults was found to have no impact on the expression of AT, AstA, AstB, AstCC, AstAR, and AstBR. However, the expression of AstCCC and AstCR was significantly upregulated in dsGlucl α -injected insects (Figure 5-figure supplement 2A-H). Further studies are required to delineate the direct or indirect mechanisms underlying this effect of Glucl α -knockdown.” (line 643-649). And we have removed Figure 5J in the revised manuscript.

(2) Expanded Discussion of Alternative Mechanisms

In the Discussion section, we have incorporated a dedicated paragraph to explore alternative pathways and compensatory mechanisms:

Key additions:

“This EB action on GluCl α expression is likely indirect, and we do not consider EB as transcriptional regulator of GluCl α . Thus, the mechanism behind EB-mediated induction of GluCl α remains to be determined. It is possible that prolonged EB exposure triggers feedback mechanisms (e.g. cellular stress responses) to counteract EB-induced GluCl α dysfunction, leading to transcriptional upregulation of the channel. Hence, considering that EB exposure in our experiments lasts several days, these findings might represent indirect (or secondary) effects caused by other factors downstream of GluCl signaling that affect channel expression.” (line 837-845).

(2) In the response to reviewers, the authors have mentioned line numbers in the main text where changes were made. But very frequently, those lines do not refer to the changes or mention just a subsection of changes done. As an example please see point 1 of Specific Points below. The problem is throughout the document making it very difficult to follow the revision and contributing to the point mentioned above.

Thank you for highlighting this critical oversight. We sincerely apologize for the inconsistency in referencing line numbers and incomplete descriptions of revisions, which undoubtedly hindered your ability to track changes effectively. We have eliminated all vague or incomplete line number references from the response letter. Instead, revisions are now explicitly tied to specific sections, figures, or paragraphs.

(3) The authors need to infer the performed experiments rationally without over interpretation. Currently, many of the claims that the authors are making are unsubstantiated. As a result of the first review process, the authors have acknowledged the discrepancies, but they have failed to alter their interpretations accordingly.

We fully agree that overinterpretation of data undermines scientific rigor. In response to your feedback, we have systematically revised the manuscript to align claims strictly with experimental evidence and to eliminate unsubstantiated assertions. We sincerely apologize for the earlier overinterpretations and appreciate your insistence on precision. The revised manuscript now rigorously distinguishes between observations (e.g., EB-GluCl-JH correlations) and hypotheses (e.g., GluCl's mechanistic role). By tempering causal language and integrating competing explanations, we aimed to present a more accurate and defensible narrative.

SPECIFIC POINTS (to each question initially raised and their rebuttals)

(1a) "Actually, there are many studies showing that insects treated with insecticides can increase the expression of target genes". Please note what is asked for is that the ligand itself induces the expression of its receptor. Of course, insecticide treatment will result in the changes expression of targets. Of all the evidences furnished in rebuttal, only Peng et al. 2017 fits the above definition. Even in this case, the accepted mode of action of chlorantraniliprole is by inducing structural change in ryanodine receptor. The observed induction of ryanodine receptor chlorantraniliprole can best be described as secondary effect. All others references do not really suffice the point asked for.

We appreciate the reviewers' suggestions for improving the manuscript. First, we have supplemented additional studies supporting the notion that " There are several studies showing that insects treated with insecticides display increases in the expression of target genes. For example, the relative expression level of the ryanodine receptor gene of the rice stem borer, *Chilo suppressalis* was increased 10-fold after treatment with chlorantraniliprole, an insecticide which targets the ryanodine receptor (Peng et al., 2017). In *Drosophila*, starvation (and low insulin) elevates the transcription level of the receptors of the neuropeptides short neuropeptide F and tachykinin (Ko et al., 2015; Root et al., 2011). In BPH,

reduction in mRNA and protein expression of a nicotinic acetylcholine receptor $\alpha 8$ subunit is associated with resistance to imidacloprid (Zhang et al., 2015). Knockdown of the $\alpha 8$ gene by RNA interference decreased the sensitivity of *N. lugens* to imidacloprid (Zhang et al., 2015). Hence, the expression of receptor genes may be regulated by diverse factors, including insecticide exposure.” We have inserted text in lines 846-857 to elaborate on these possibilities.

Second, we would like to reiterate our position: we have merely described this phenomenon, specifically that EB treatment increases GluCl α expression. “This EB action on GluCl α expression is likely indirect, and we do not consider EB as transcriptional regulator of GluCl α . Thus, the mechanism behind EB-mediated induction of GluCl α remains to be determined. It is possible that prolonged EB exposure triggers feedback mechanisms (e.g. cellular stress responses) to counteract EB-induced GluCl α dysfunction, leading to transcriptional upregulation of the channel. Hence, considering that EB exposure in our experiments lasts several days, these findings might represent indirect (or secondary) effects caused by other factors downstream of GluCl signaling that affect channel expression.” We have inserted text in lines 837-845 to elaborate on these possibilities.

Once again, we sincerely appreciate this discussion, which has provided us with a deeper understanding of this phenomenon.

b. The authors in their rebuttal accepts that they do not consider EB to a transcriptional regulator of Glucl α and the induction of Glucl α as a result of EB can best be considered as a secondary effect. But that is not reflected in the manuscript, particularly in the result section. Current state of writing implies EB up regulation of Glucl α to an important event that contributes majorly to the hypothesis. So much so that they have retained the schematic diagram (Fig. 5J) where EB $\rightarrow$ Glucl α is drawn. Even the heading of the subsection says "EB-enhanced fecundity in BPHs is dependent on its molecular target protein, the Glucl α channel". As mentioned in the general points, it is not enough to have a good rebuttal written to the reviewer, the parent manuscript needs to reflect on the changes asked for.

Thank you for your comments. We have carefully addressed your suggestions and made corresponding revisions to the manuscript.

We fully acknowledge the reviewer's valid concern. In this revised manuscript, “However, we do not propose that EB is a direct transcriptional regulator of Glucl α , since EB and other avermectins are known to alter the channel conformation and thus their function (Wolstenholme, 2012; Wu et al., 2017). Thus, it is likely that the observed increase in Glucl α transcript is a secondary effect downstream of EB signaling.” (Line 625-629). We agree that the original presentation in the manuscript, particularly within the Results section, did not adequately reflect this nuance and could be misinterpreted as suggesting a direct regulatory role for EB on Glucl α transcription.

Regarding Fig. 5J, we have removed the figure and all mentions of Fig. 5J and its legend in the revised manuscript.

c. "We have inserted text on lines 738 - 757 to explain these possibilities." Not a single line in the section mentioned above discussed the topic in hand. This is serious undermining of the review process or carelessness to the extreme level.

In the Results section, we have now added descriptions “Taken together, these results reveal that EB exposure is associated with an increase in JH titer and that this elevated JH signaling contributes to enhanced fecundity in BPH.” (line 375-377).

For the figures, we have removed Fig. 4N and all mentions of Fig. 4N and its legend in the revised manuscript.

Lastly, regarding the issue of locating specific lines, we deeply regret any inconvenience caused. Due to the track changes mode used during revisions, line numbers may have shifted, resulting in incorrect references. We sincerely apologize for this and have now corrected the line numbers.

(2) The section written in rebuttal should be included in the discussion as well, explaining why authors think a nymphal treatment with JH may work in increasing fecundity of the adults. Also, the authors accept that EBs effect on JH titer in Indirect. The text of the manuscript, results section and figures should be reflective of that. It is NOT ok to accept that EB impacts JH titer indirectly in a rebuttal letter while still continuing to portray EB direct effect on JH titer. In terms of diagrams, authors cannot put a -> sign until and unless the effect is direct. This is an accepted norm in biological publications.

We appreciate the reviewer's valuable suggestions here. We have now carefully revised the manuscript to address all concerns, particularly regarding the mechanism linking nymphal EB exposure to adult fecundity and the indirect nature of EB's effect on JH titers. Below are our point-by-point responses and corresponding manuscript changes. Revised text is clearly marked in the resubmitted manuscript.

(1) Clarifying the mechanism linking nymphal EB treatment to adult fecundity:

Reviewer concern: Explain why nymphal EB treatment increases adult fecundity despite undetectable EB residues in adults.

Response & Actions Taken:

We agree this requires explicit discussion. We now propose that nymphal EB exposure triggers developmental reprogramming (e.g., metabolic/epigenetic changes) that persist into adulthood, indirectly enhancing JH synthesis and fecundity. This is supported by two key findings:

(1) No detectable EB residues in adults after nymphal treatment (new Figure 1–figure supplement 1C).

(2) Increased adult weight and nutrient reserves (Figure 1–figure supplement 3E,F), suggesting altered resource allocation.

Added to Discussion (Lines 793–803): Notably, after exposing fourth-instar BPH nymphs to EB, no EB residues were detected in the subsequent adult stage. This finding indicates that the EB-induced increase in adult fecundity is initiated during the nymphal stage and manifests in adulthood - a mechanism distinct from the direct fecundity enhancement of fecundity observed when EB is applied to adults. We propose that sublethal EB exposure during critical nymphal stages may reprogram metabolic or endocrine pathways, potentially via insulin/JH crosstalk. For instance, increased nutrient storage (e.g., proteins, sugars; Figure 2–figure supplement 2) could enhance insulin signaling, which in turn promotes JH biosynthesis in adults (Ling and Raikhel, 2021; Mirth et al., 2014; Sheng et al., 2011). Future studies should test whether EB alters insulin-like peptide expression or signaling during development.

(3) Emphasizing EB's indirect effect on JH titers: Reviewer concern: The manuscript overstated EB's direct effect on JH. Arrows in figures implied causality where only correlation exists.

Response & Actions

Taken: We fully agree. EB's effect on JH is indirect and multifactorial (via AstA/AstAR suppression, GluCl modulation, and metabolic changes). We have:

Removed oversimplified schematics (original Figures 3N, 4N, 5J).

Revised all causal language (e.g., "EB increases JH" → "EB exposure is associated with increased circulating JH III "). (Line 739)

Clarified in Results/Discussion that EB-induced JH changes are likely secondary to neuroendocrine disruption.

Key revisions:

Results (Lines 375–377):

"Taken together, these results reveal that EB exposure is associated with an increase in JH titer and that JH signaling contributes to enhanced fecundity in BPH."

Discussion (Lines 837–845):

This EB action on GluCl α expression is likely indirect, and we do not consider EB as transcriptional regulator of GluCl α . Thus, the mechanism behind EB-mediated induction of GluCl α remains to be determined. It is possible that prolonged EB exposure triggers feedback mechanisms (e.g. cellular stress responses) to counteract EB-induced GluCl α dysfunction, leading to transcriptional upregulation of the channel. Hence, considering that EB exposure in our experiments lasts several days, these findings might represent indirect (or secondary) effects caused by other factors downstream of GluCl signaling that affect channel expression.

| *a. Lines 281-285 as mentioned, does not carry the relevant information.*

Thank you for your careful review of our manuscript. We sincerely apologize for the confusion regarding line references in our previous response. Due to extensive revisions and tracked changes during the revision process, the line numbers shifted, resulting in incorrect citations for Lines 281–285. The correct location for the added results (EB-induced increase in mature eggs in adult ovaries) is now in lines 253-258: "We furthermore observed that EB treatment of female adults also increases the number of mature eggs in the ovary (Figure 2-figure supplement 1)."

| *b. Lines 351-356 as mentioned, does not carry the relevant information. Lines 281-285 as mentioned, does not carry the relevant information.*

Thank you for your careful review of our manuscript. We sincerely apologize for the confusion regarding line references in our previous response. The correct location for the added results is now in lines 366-371: "We also investigated the effects of EB treatment on the JH titer of female adults. The data indicate that the JH titer was also significantly increased in the EB-treated female adults compared with controls (Figure 3-figure supplement 3A). However, again the steroid 20-hydroxyecdysone, was not significantly different between EB-treated BPH and controls (Figure 3-figure supplement 3B)."

| *c. Lines 378-379 as mentioned, does not carry the relevant information. Lines 387-390 as mentioned, does not carry the relevant information.*

We sincerely apologize for the confusion regarding line references in our previous response.

The correct location for the added results is now in lines 393-394: We furthermore found that EB treatment in female adults increases JHAMT expression (Figure 3-figure supplement 3C).

The other correct location for the added results is now in lines 405-408: We found that Kr-h1 was significantly upregulated in the adults of EB-treated BPH at the 5M, 5L nymph and 4 to 5 DAE stages (4.7-fold to 27.2-fold) when 4th instar nymph or female adults were treated with EB (Figure 3H and Figure 3-figure supplement 3D)..

(3) The writing quality is still extremely poor. It does not meet any publication standard, let alone elife.

We fully understand your concerns and frustrations, and we sincerely apologize for the deficiencies in our writing quality, which did not meet the high standards expected by you and the journal. We fully accept your criticism regarding the writing quality and have rigorously revised the manuscript according to your suggestions.

(4) I am confused whether Figure 2B was redone or just edited. Otherwise this seems acceptable to me.

Regarding Fig. 2B, we have edited the text on the y-axis. The previous wording included the term “retention,” which may have caused misunderstanding for both the readers and yourself, leading to the perception of contradiction. We have now revised this wording to ensure accurate comprehension.

(5) The rebuttal is accepted. However, still some of the lines mentioned does not hold relevant information.

This error has been corrected.

The correct location for the added results is now in lines 255-258 and lines 279-282: “Hence, although EB does not affect the normal egg developmental stages (see description in next section), our results suggest that EB treatment promotes oogenesis and, as a result the insects both produce more eggs in the ovary and a larger number of eggs are laid.” and “However, considering that the number of eggs laid by EB treated females was larger than in control females (Figure 1 and Figure 1-figure supplement 1), our data indicates that EB treatment of BPH can both promote both oogenesis and oviposition.”

(6) Thank you for the clarification. Although now discussed extensively in discussion section, the nuances of indirect effect and minimal change in expression should also be reflected in the result section text. This is to ensure that readers have clear idea about content of the paper.

Corrected. To ensure readers gain a clear understanding of our data, we have briefly presented these discussions in the Results section. Please see line 397-402: The levels of met mRNA slightly increased in EB-treated BPH at the 5M and 5L instar nymph and 1 to 5 DAE adult stages compared to controls (1.7-fold to 2.9-fold) (Figure 3G). However, it should be mentioned that JH action does not result in an increase of Met. Thus, it is possible that other factors (indirect effects), induced by EB treatment cause the increase in the mRNA expression level of Met.

(7) As per the author's interpretation, it becomes critical to quantitate the amount of EB present at the adult stages after a 4th instar exposure to it. Only this experiment will unambiguously proof the authors claim. Also, since they have done adult insect exposure to EB, such experiments should be systematically performed for as many sections as possible. Don't just focus on few instances where reviewers have pointed out the issue.

Thank you for raising this critical point. To address this concern, we have conducted new supplementary experiments. The new experimental results demonstrate that residual levels of emamectin benzoate (EB) in adult-stage brown planthoppers (BPH) were below the instrument detection limit following treatment of 4th instar nymphs with EB. Line 172-184: “To determine whether EB administered during the fourth-instar larval stage persists as residues in the adult stage, we used HPLC-MS/MS to quantify the amount of EB present at the adult stage after exposing 4th-instar nymphs to this compound. However, we found no detectable EB residues in the adult stage following fourth-instar nymphal treatment (Figure 1-figure supplement 1C). This suggests that the mechanism underlying the increased fecundity of female adults induced by EB treatment of nymphs may differ from that caused by direct EB treatment of female adults. Combined with our previous observation that EB treatment significantly increased the body weight of adult females (Figure 1—figure supplement 3E and F), a possible explanation for this phenomenon is that EB may enhance food intake in BPH, potentially leading to elevated production of insulin-like peptides and thus increased growth. Increased insulin signaling could potentially also stimulate juvenile hormone (JH) biosynthesis during the adult stage (Badisco et al., 2013).”

(8) Thank you for the revision. Lines 725-735 as mentioned, does not carry the relevant information. However, since the authors have decided to remove this systematically from the manuscript, discussion on this may not be required.

Thank you for identifying the limited relevance of the content in Lines 725–735 of the original manuscript. As recommended, we have removed this section in the revised version to improve logical coherence and maintain focus on the core findings.

(9) Normally, dsRNA would last for some time in the insect system and would down-regulate any further induction of target genes by EB. I suggest the authors to measure the level of the target genes by qPCR in KD insects before and after EB treatment to clear the confusion and unambiguously demonstrate the results. Please Note- such quantifications should be done for all the KD+EB experiments. Additionally, citing few papers where such a rescue effect has been demonstrated in closely related insect will help in building confidence.

We appreciate the reviewer’s suggestion to clarify the interaction between RNAi-mediated gene knockdown (KD) and EB treatment. To address this, we performed additional experiments measuring *Kr-h1* expression via qPCR in *dsKr-h1*-injected insects before and after EB exposure.

The results (now Figure 3—figure supplement 4) show that:

- (1) EB did not rescue **Kr-h1** suppression at 24h post-treatment (**p** > 0.05).
- (2) Partial recovery of fecundity occurred later (Figure 3M), likely due to:
 - a) Degradation of dsRNA over time, reducing KD efficacy (Liu et al., 2010).
 - b) Indirect effects of EB (e.g., hormonal/metabolic reprogramming) compensating for residual *Kr-h1* suppression.

Please see line 441-453: “Next, we investigated whether EB treatment could rescue the dsRNA-mediated gene silencing effect. To address this, we selected the *Kr-h1* gene and analyzed its expression levels after EB treatment. Our results showed that *Kr-h1* expression was suppressed by ~70% at 72 h post-dsRNA injection. However, EB treatment did not significantly rescue *Kr-h1* expression in gene knock down insects (**p** > 0.05) at 24h post-EB treatment (Figure 3-figure supplement 4). While dsRNA-mediated *Kr-h1* suppression was

robust initially, its efficacy may decline during prolonged experiments. This aligns with reports in BPH, where effects of RNAi gradually diminish beyond 7 days post-injection (Liu et al., 2010a). The late-phase fecundity increase might reflect partial Kr-h1 recovery due to RNAi degradation, allowing residual EB to weakly stimulate reproduction. In addition, the physiological impact of EB (e.g., neurotoxicity, hormonal modulation) could manifest via compensatory feedback loops or metabolic remodeling.”

(10) Not a very convincing argument. Besides without a scale bar, it is hard for the reviewers to judge the size of the organism. Whole body measurements of JH synthesis enzymes will remain as a quite a drawback for the paper.

In response to your suggestion, we have also included images with scale bars (see next Figure 1). The images show that the head region is difficult to separate from the brown thoracic sclerite region. Furthermore, the anatomical position of the Corpora Allata in brown planthoppers has never been reported, making dissection uncertain and highly challenging. To address this, we are now attempting to use *Drosophila* as a model to investigate how EB regulates JH synthesis and reproduction.

Author response image 1.

This illustration provides a visual representation of the brown planthopper (BPH), a major rice pest.



Figure 1. This illustration provides a visual representation of the brown planthopper (BPH), a major rice pest.).

(11) "The phenomenon reported was specific to BPH and not found in other insects. This limits the implications of the study". This argument still holds. Combined with extreme species specificity, the general effect that EB causes brings into question the molecular specificity that the authors claim about the mode of action.

We acknowledge that the specificity of the phenomenon to BPH may limit its broader implications, but we would like to emphasize that this study provides important insights into the unique biological mechanisms in BPH, a pest of significant agricultural importance. The molecular specificity we described in the manuscript is based on rigorous experimental evidence. We believe that it contributes to valuable knowledge to understand the interaction of external factors such as EB and BPH and resurgence of pests. We hope that this study will inspire further research into the mechanisms underlying similar phenomena in other insects, thereby broadening our understanding of insect biology. Since EB also has an effect on fecundity in *Drosophila*, albeit opposite to that in BPHs (Fig. 1 suppl. 2), it seems likely that EB actions may be of more general interest in insect reproduction.

(12) The authors have added a few lines in the discussion but it does not change the overall design of the experiments. In this scenario, they should infer the performed experiments rationally without over interpretation. Currently, many of the claims that the authors are making are unsubstantiated. As a result of the first review process, the authors have acknowledged the discrepancies, but they have failed to alter their interpretations accordingly.

We appreciate your concern regarding the experimental design and the need for rational inference without overinterpretation. In response, we would like to clarify that our discussion is based on the experimental data we have collected. We acknowledge that our study focuses on BPH and the specific effects of EB, and while we agree that broader generalizations require further research, we believe the new findings we present are valid and contribute to the understanding of this specific system.

We also acknowledge the discrepancies you mentioned and have carefully considered your suggestions. In this revised version, we believe our interpretations are reasonable and consistent with the data, and we have adjusted our discussion to better reflect the scope of our findings. We hope that these revisions address your concerns. Thank you again for your constructive feedback.

ADDITIONAL POINTS

(1) Only one experiment was performed with Abamectin. No titration for the dosage were done for this compound, or at least not provided in the manuscript. Inclusion of this result will confuse readers. While removing this result does not impact the manuscript at all. My suggestion would be to remove this result.

We acknowledge that the abamectin experiment lacks dose-titration details and that its standalone presentation could lead to confusion. However, we respectfully request to retain these results for the following reasons:

Class-Specific Mechanism Validation:

Abamectin and emamectin benzoate (EB) are both macrocyclic lactones targeting glutamate-gated chloride channels (GluCl_s). The observed similarity in their effects on BPH fecundity (e.g., Figure 1—figure supplement 1B) supports the hypothesis that GluCl modulation, rather than compound-specific off-target effects, drives the reproductive enhancement. This consistency strengthens the mechanistic argument central to our study.

(2) The section "The impact of EB treatment on BPH reproductive fitness" is poorly described. This needs elaboration. A line or two should be included to describe why the parameters chosen to decide reproductive fitness were selected in the first place. I see that the definition of brachypterism has undergone a change from the first version of the manuscript. Can you provide an explanation for that? Also, there is no rationale behind

inclusion of statements on insulin at this stage. The authors have not investigated insulin. Including that here will confuse readers. This can be added in the discussion though.

Thank you for your suggestion. We have added an explanation regarding the primary consideration of evaluating reproductive fitness. In the interaction between sublethal doses of insecticides and pests, reproductive fitness is a key factor, as it accurately reflects the potential impact of insecticides on pest control in the field. Among the reproductive fitness parameters, factors such as female *Nilaparvata lugens* body weight, lifespan, and brachypterous ratio (as short-winged *N. lugens* exhibit higher oviposition rates than long-winged individuals) are critical determinants of reproductive success. Therefore, we comprehensively assessed the effects of EB on these parameters to elucidate the primary mechanism by which EB influences reproduction. We sincerely appreciate your constructive feedback.

(3) "EB promotes ovarian maturation in BPH" this entire section needs to be rewritten and attention should be paid to the sequence of experiments described.

Thank you for your suggestion. Based on your recommendation, we have rewritten this section (lines 267–275) and adjusted the sequence of experimental descriptions to improve the structural clarity of this part.

(4) Figure 3N is outright wrong and should be removed or revised.

In accordance with your recommendation, we have removed the figure.

(5) When you are measuring hormonal titers, it is important to mention explicitly whether you are measuring hemolymph titer or whole body.

We believe we have explicitly stated in the Methods section (line 1013) that we measured whole-body hormone titers. However, we now added this information to figure legends.

(6) EB induces JH biosynthesis through the peptidergic AstA/AstAR signaling pathway- this section needs attention at multiple points. Please check.

We acknowledge that direct evidence for EB-AstA/AstAR interaction is limited and have framed these findings as a hypothesis for future validation.

References

Liu, S., Ding, Z., Zhang, C., Yang, B., Liu, Z., 2010. Gene knockdown by intro-thoracic injection of double-stranded RNA in the brown planthopper, *Nilaparvata lugens*. *Insect Biochem. Mol. Biol.* 40, 666-671

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